

Contents

Introduction	1
1 General background	3
1.1 Organization of the genome	3
1.1.1 DNA	4
1.1.2 Chromatin	6
1.2 Gene transcription	6
1.2.1 RNA	7
1.2.2 RNA polymerase and transcriptional machinery.	8
2 Transcription factor-DNA interactions	11
2.1 Cis-regulatory regions.	11
2.1.1 Promoters	11
2.1.2 Enhancers and silencers	13
2.2 Transcription factors	16
2.2.1 Structural classification	16
2.2.2 Functional classification	18
2.3 Transcription factor binding to the DNA.	19
2.3.1 Direct and indirect TF binding	19
2.3.2 Cooperative transcription factor binding	20
2.3.3 Impact of methylation to the DNA binding	21
2.3.4 Experimental methods for TF-DNA interaction detection	23
3 Computational modeling of transcription factor-DNA interactions	31
3.1 Transcription factor DNA binding models	31
3.1.1 Position weight matrices (PWMs).	31
3.1.2 Other more complex models.	32
3.1.3 <i>De novo</i> motif discovery	36
3.1.4 Non-negative matrix factorization for motif discovery.	37
3.2 Databases of TF-DNA interaction models.	37
3.3 Use of TF-DNA binding models	39
3.3.1 Prediction of TF binding sites	39
3.3.2 Prediction of direct TF binding sites.	40
3.3.3 Annotating <i>de novo</i> discovered motifs.	41
3.3.4 Prediction of TF co-binding events	41

Summary of papers	43
4 Summary of papers	45
4.1 Paper I. JASPAR 2022: the 9th release of the open-access database of transcription factor binding profiles	45
4.2 Paper II. JASPAR 2024: 20th anniversary of the open-access database of transcription factor binding profiles	46
4.3 Paper III. Identification of transcription factor co-binding patterns with non-negative matrix factorization	48
Discussion	51
5 Discussion	53
5.1 The current status and future perspectives of computational models for TF-DNA interactions.	53
5.2 Identifying and defining TF co-binding	55
5.3 The need for maintenance and update of the scientific databases	57
5.4 Development of new bioinformatics tools	59
Bibliography	63
Papers	77
I JASPAR 2022: the 9th release of the open-access database of transcription factor binding profiles	79
II JASPAR 2024: 20th anniversary of the open-access database of transcription factor binding profiles	91
III Identification of transcription factor co-binding patterns with non-negative matrix factorization	103

List of Figures

1.1	Eukaryotic genome organization	4
1.2	DNA structure	5
1.3	RNA structure	7
1.4	From gene to protein.	9
2.5	Regulatory regions and transcription regulation.	12
2.6	Main classes of promoters	14
2.7	Chromatin loop and TAD formation	15
2.8	TF-DNA binding and TF classification.	17
2.9	TF cooperativity binding	21
2.10	TF cooperativity models	22
2.11	TF binding to methylated DNA.	24
2.12	Experimental methods to investigate TF-DNA interactions	26
2.13	Single-molecule footprinting.	29
3.14	TF-DNA binding computational modelling	34
3.15	Deep learning modeling approaches	35
3.16	NMF for DNA pattern discovery	38
3.17	Use of TF-DNA binding models to infer direct TF-DNA interactions.	42

List of Figures

Preface

This thesis is submitted in partial fulfillment of the requirements for the degree of *Philosophiae Doctor* at the University of Oslo. The presented work was conducted at the Centre for Molecular Medicine Norway (NCMM), organized by the University of Oslo, under the supervision of Professor Dr. Anthony Mathelier and the co-supervision of Associate Professor Dr. Jonas Paulsen. The Research Council of Norway funded this research through grants 187615 and 288404, the Norwegian Cancer Society through grant 197884, and the Nordic EMBL Partnership Hub for Molecular Medicine through NordForsk grant 96782.

A collection of three articles is the basis of this thesis. Two papers were published, and one is in peer review at the thesis submission date. The overarching theme of the presented articles is transcription factor binding to the DNA. Introductory chapters provide the necessary theoretical overview and motivation for this research.

Acknowledgements

Preface

Abbreviations

3'UTR	three prime untranslated region
5'UTR	five prime untranslated region
5mC	5-methylcytosine
A	adenine
ATAC-seq	assay for transposase-accessible chromatin using sequencing
bHLH	basic helix-loop-helix
bp	base pair
C	cytosine
CAP-SELEX	consecutive affinity-purification systematic evolution of ligands by exponential enrichment
CGI	CpG island
ChIP-seq	chromatin immunoprecipitation followed by sequencing
CNN	convolutional neural network
CRRs	cis-regulatory regions
DAP-seq	DNA affinity purification sequencing
DBD	DNA-binding domain
DHS	DNase I hypersensitivity
DNA	deoxyribonucleic acid
DNMTs	DNA methyltransferases
DOI	Digital Object Identifier
eRNA	enhancer RNA
FAIR	Findable, Accessible, Interoperable, Reusable
G	guanine

Preface

GTFs	general transcription factors
H3K27ac	Histone 3 Lysine 27 acetylation
H3K27me3	Histone 3 Lysine 27 trimethylation
H3K36me	Histone 3 Lysine 36 methylation
H3K4me3	Histone 3 Lysine 4 trimethylation
H3K9me3	Histone 3 Lysine 9 trimethylation
H4K20me	Histone 4 Lysine 20 methylation
HMMs	hidden Markov models
IC	information content
kb	(or kbp) kilobase pair
lncRNAs	long non-coding RNAs
Mb	(or Mbp) megabase pair
mESCs	mouse embryonic stem cells
miRNAs	micro RNAs
mRNA	messenger RNA
ncRNAs	non-coding RNAs
NMF	non-negative matrix factorization
NTPs	nucleotide triphosphates
ORF	open reading frame
PBM	protein binding microarray
PDB	Protein Data Bank
PFM	position frequency matrix
PIC	preinitiation complex
piRNAs	PIWI-interacting RNAs
Pol II	RNA polymerase II
PPI	protein-protein interaction
PPM	position probability matrix
pre-mRNA	precursor mRNA

PSSMs	position specific scoring matrices
PTMs	post-translational modifications
PWMs	position weight matrices
RNA	ribonucleic acid
rRNA	ribosomal RNA
SELEX	systematic evolution of ligands by exponential enrichment
siRNAs	small-interfering RNAs
SMF	single-molecule footprinting
snoRNAs	small nucleolar RNAs
snRNAs	small nuclear RNAs
T	thymine
TADs	topologically associated domains
TAFs	TBP-associated factors
TBP	TATA-binding protein
TET	ten eleven translocation
TFBSs	transcription factor binding sites
TFFMs	transcription factor flexible models
TFs	transcription factors
tRNA	transfer RNA
TSSs	transcription start sites
U	uracil

Preface

Introduction

Chapter 1

General background

Humans are multicellular organisms composed of hundreds of cell types. Every cell in an organism has deoxyribonucleic acid (DNA) stored in the cell's nucleus. DNA is the hereditary material that contains the instructions for a cell to perform its functions, divide, and differentiate. The cells in an organism must also communicate with each other and the environment to coordinate all the processes. Four bases, adenine (A), thymine (T), cytosine (C), and guanine (G), are composing DNA sequences (genetic code) to encode a blueprint of life. All genetic material within a single cell is known as a genome. Specific sequences within DNA, termed genes, are considered functional units of heredity. A subset of genes encode proteins (protein-coding), while other genes do not (non-coding). Proteins are molecules that perform most of the cellular functions. A gene is first copied (transcribed) into ribonucleic acid (RNA) during transcription. In the case of protein-coding genes, RNA molecules are then used in translation to inform what amino acids should be brought and connected into a chain to form a protein. RNA molecules transcribed from non-coding genes are further processed and participate in other cellular processes, such as transcription regulation.

Every cell in an organism has the same set of genes. However, cells are different and form different tissues and organs. What enables cells to differentiate into many types when the DNA sequences are identical? Gene transcription is highly regulated, ensuring that specific cells express the right genes and gene combinations at the appropriate times and locations. Major players in this process are a particular class of proteins — transcription factors (TFs). They act by binding to specific DNA sequences (regulatory regions), which can control the associated gene transcription rate. Interactions between TFs and regulatory DNA sequences ensure gene transcription control in space and time. This chapter overviews basic molecular biology concepts, such as DNA, RNA, genome, and the process of transcription, to provide the background to dive deeper into how transcription is regulated.

1.1 Organization of the genome

In eukaryotes, the cell nucleus stores hereditary information as DNA. Cells copy (replicate) genomic DNA during division or transcribe portions of the DNA (genes)

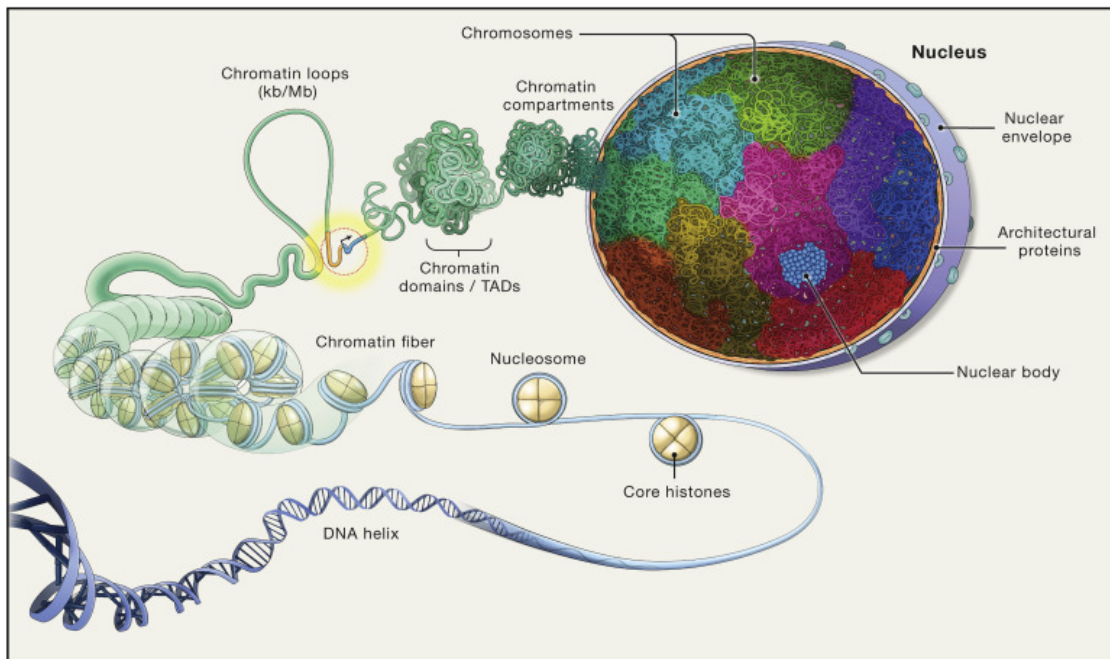


Figure 1.1: Eukaryotic genome organization. Genomic DNA is wrapped around histones to form nucleosomes, the basic units of chromatin fiber. Chromatin can form loops that span kilobase pairs (kb) or megabase pairs (Mb) and are essential for gene expression regulation. Bigger stretches of chromatin are organized into chromatin domains or topologically associated domains (TADs), forming chromatin compartments to pack everything into the nucleus. Each chromosome occupies a territory in the nucleus. The nucleus also contains a nuclear body and architectural proteins. A nuclear envelope surrounds the contents of the nucleus. Adapted from [2].

into RNA to produce proteins. Human DNA is only 2 nanometers wide, but if we stretch it into a single thread, it would be around 2 meters long [1]. The microscopic cell nucleus stores tightly compacted DNA. Multiple levels of genome organization are applied. The simplest level of organization is a DNA helix wrapped around a specific class of proteins called histones. The highest level of organization is a chromosome, visible during the cell division. A chromosome consists of a single molecule of DNA and various proteins. The human genome is organized into 22 pairs of autosomal chromosomes and one pair of sex chromosomes (XX or XY). During interphase, DNA organizes into chromosome territories in the nucleus (Figure 1.1) [2]. In addition to the DNA and architectural proteins, the nucleus contains different types of nuclear bodies (nucleolus, paraspeckles), made of proteins and involved in cellular processes, such as telomere maintenance or stress response [2, 3]. A nuclear envelope surrounds the content of the nucleus. The specific organization ensures proper genome function — expressing genes when and where needed.

1.1.1 DNA

The existence of DNA was discovered in 1869 by Friedrich Miescher when he found a new substance inside the nuclei of the human white blood cells that did not have

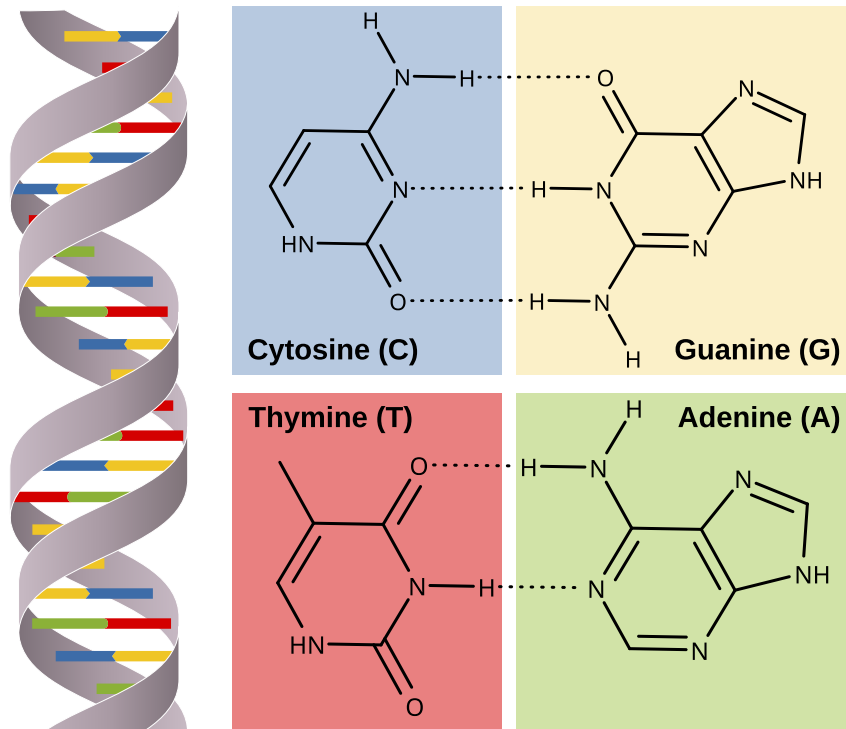


Figure 1.2: DNA structure. The two chains of the DNA are composed of a sugar backbone and four nucleotides: adenine (A), thymine (T), cytosine (C), and guanine (G). Nucleotides in the two chains connect via hydrogen bonds in a strict complementarity, where A is bound to T and C is bound to G.

the same properties as proteins or lipids [4, 5]. In 1953, contributions of Rosalind Franklin, James Watson, and Francis Crick led to the publication of resolved DNA molecule structure and is often considered the beginning of molecular biology [6]. This fundamental moment in science history allowed us to explain the possible mechanism of DNA replication, thus information multiplication and transfer.

DNA stores genetic information in the cell. In the DNA structure, the sugar backbone (deoxyribose) holds four main types of nucleotides: purines adenine (A) and guanine (G), and pyrimidines thymine (T) and cytosine (C) [7] [1]. Chargaff's rule states that in the DNA of any species, the proportions of A and T nucleotides and C and G are always equal, demonstrating unique base-pairing (complementarity), when forming a double-stranded DNA helix [7, 8]. A-T and C-G nucleotide pairs in two strands of the DNA and are held by two and three hydrogen bonds, respectively (Figure 1.2).

The most commonly found DNA structure is called a B form of DNA. Two helical DNA strands entwine to form spaces between the strands, major and minor grooves, which are 22 and 12 Å wide, respectively. The grooves create space for the proteins to interact with the double-stranded DNA. One of the most important classes of proteins that bind DNA is transcription factors (TFs) [9]. TFs are essential for the regulation of transcription. They act by recognizing specific sequences of nucleotides.

1.1.2 Chromatin

Chromatin is a mixture of DNA and proteins that form chromosomes, the highest level of genome organization. The fundamental element of chromatin is a nucleosome, which consists of 146 bp of DNA wrapped around the core histone protein octamer. Nucleosomes gather into a chromatin fiber typically 5-24 nanometers in diameter. The fibers organize into higher structures. They fold into chromatin domains, called topologically associated domains (TADs) [10]. TADs associate with each other and create chromatin compartments. When cells are dividing, chromatin assembles into a metaphase chromosome. During interphase, chromatin is organized into domains and compartments and tends to localize in specific territories (Figure 1.1) [2, 11]. Chromatin fibers from different chromosomes interact at the edges but do not entangle. In addition to chromatin, the nucleus contains protein structures called nuclear bodies. Nuclear bodies, such as nucleolus, Cajal bodies, paraspeckles, and others, are membrane-less organelles [12]. Their functions include the coordination and processing of RNAs and coordination of chromatin-modifying proteins, among others [12]. The nuclear envelope surrounds the contents of the nucleus.

Canonical core histone proteins are H2A, H2B, H3, and H4, with two copies each in a single octameric histone core (Figure 1.1). A linker histone H1 separates every core nucleosome [11, 13]. The variants in specific genomic regions replace well-conserved canonical histone proteins. For example, H3.3 replaces the H3 histone in active genes and promoter regions [13, 14]. Histone protein N-terminal tails undergo extensive post-translational modifications (PTMs), such as acetylation and methylation, on their amino acids. These modifications are essential for controlling the accessibility of the chromatin. Accessible chromatin is accessed by a transcriptional machinery and regulatory proteins [13].

Based on its activity state, the chromatin can be heterochromatin or euchromatin [15]. Euchromatin is a more open, active chromatin with more widely spaced nucleosomes. Several histone tail chemical modifications (Histone 3 Lysine 4 trimethylation (H3K4me3), Histone 3 Lysine 27 acetylation (H3K27ac), Histone 3 Lysine 36 methylation (H3K36me)) and histone variants (H3.3, H2A.Z, H2A.B) are marks of active transcription. Heterochromatin is generally known as a silent chromatin and is less accessible and transcribed. Constitutive heterochromatin is more static and contains Histone 3 Lysine 9 trimethylation (H3K9me3). Histone 3 Lysine 27 trimethylation (H3K27me3) marks the facultative heterochromatin, which can switch between active and inactive states. For example, facultative heterochromatin marks an inactive X-chromosome [11].

1.2 Gene transcription

Genes represent basic units of heredity. A gene is a stretch of DNA in the genome subjected to transcription by RNA polymerase. Transcription is a process of copying DNA and producing an RNA molecule. Genes can be protein-coding or non-coding. The protein-coding RNA template is translated into a protein, while

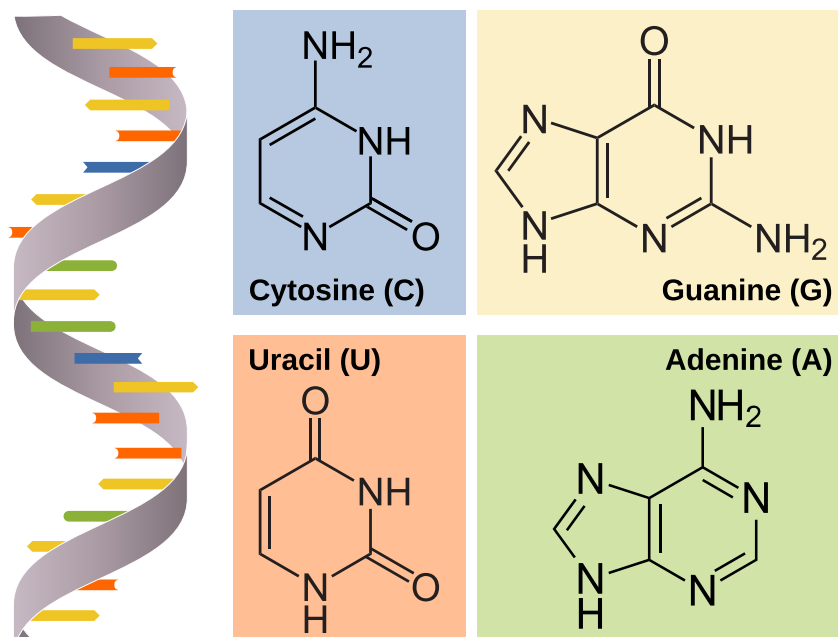


Figure 1.3: RNA structure. RNA is a single-stranded molecule composed of a sugar backbone and four nucleotides: uracil (U), adenine (A), cytosine (C), and guanine (G).

non-coding genes produce non-coding RNA molecules. A collection of proteins assembles into transcriptional machinery and organizes the transcription process.

1.2.1 RNA

The transcription machinery, with the leading player RNA polymerase, transcribes a portion of the genome to produce RNA. An RNA molecule is single-stranded and consists of a sugar backbone (ribonucleose) and four types of nucleotides: uracil (U), A, C and G (Figure 1.3). RNA molecules can adopt various secondary structures by forming hydrogen bonds between nucleotides (C pairs with G, while A pairs with U) within the same molecule [16].

The main types of RNAs, based on their functions, are messenger RNA (mRNA), ribosomal RNA (rRNA), transfer RNA (tRNA), and other non-coding RNAs (ncRNAs) [16–18]. A precursor mRNA (pre-mRNA) is produced in the nucleus during transcription by copying a protein-coding gene (Figure 1.4) [17]. RNA polymerase synthesizes RNA from nucleotide triphosphates (NTPs) (ATP, GTP, CTP, and UTP, precursors of A, G, C, and U, respectively). A pre-mRNA is heavily processed to produce an mRNA that carries information to the cytoplasm, where ribosomes translate it to a polypeptide. A mRNA has an open reading frame (ORF), consisting of introns and exons. The ORF is flanked by three prime untranslated region (3'UTR) and five prime untranslated region (5'UTR) [16, 19]. During splicing, a specific protein complex, the spliceosome, processes most of the pre-mRNA by cutting out intronic (non-protein coding) gene segments to leave only exons (protein-coding) [20]. Introns can be spliced differently by choosing to leave out some exons, creating a variety of transcripts copied from the same

gene. This process is known as alternative splicing [21]. Splicing factor proteins recognize splicing signaling sequences at the intron-exon boundaries [20, 21]. In addition to splicing, a cap is added at the 5' end of the RNA, while multiple added A nucleotides at the 3' end form a poly-A tail [21]. After this, mRNA travels from the nucleus to the ribosomes in the cytoplasm. Ribosomes receive individual mRNAs and read the information encoded. Ribosomes contain rRNA, and each amino acid is brought into the ribosome by tRNAs that recognize every three successive mRNA nucleotides, termed codons [16, 18]. Codons code for amino acids.

Longer than 200 nucleotides non-coding RNAs (long non-coding RNAs (lncRNAs)) and shorter small ncRNAs are the two main size-based categories of ncRNAs. Small ncRNAs can be divided into micro RNAs (miRNAs), small nucleolar RNAs (snoRNAs), small nuclear RNAs (snRNAs), small-interfering RNAs (siRNAs) and PIWI-interacting RNAs (piRNAs) [22]. miRNAs have an essential function in eukaryotic gene regulation. They can target and silence mRNAs to prevent translating them into proteins [17]. Aberrations of the ncRNA activities can be associated with diseases such as cancer. For example, miRNA can act as a tumor suppressor and target genes responsible for cell growth. snRNAs and snoRNAs are involved in splicing. snoRNAs can also guide the modifications of rRNAs [22]. The primary function of siRNAs is silencing the genomic locus, for example, the same locus from which siRNA is derived. piRNAs play a role in silencing the transposable elements [23].

1.2.2 RNA polymerase and transcriptional machinery

In eukaryotes, RNA polymerase II (Pol II) is responsible for transcribing genes encoding mRNAs and some ncRNAs [24]. This twelve-subunit protein uses NTPs as a substrate to produce RNA molecules from a DNA template. The RNA sequence synthesis process follows nucleotide complementarity.

Pol II is not able to recognize the transcription start sites (TSSs) by itself and needs to be guided by a set of proteins known as general transcription factors (GTFs) (Figure 1.4). Pol II and GTFs form a minimal transcription machinery, and all these proteins, together with the template DNA, form the preinitiation complex (PIC) [24, 25]. The GTFs are TFIIB, TFIID, TFIIE, TFIIF, and TFIIH, and each plays a specific role in transcription initiation. They assemble one after another. Proteins TATA-binding protein (TBP) and TBP-associated factors (TAFs) compose TFIID and recognize a particular sequence of DNA near some TSSs, called a TATA box. TFIIB and the non-phosphorylated form of Pol II join the complex, followed by TFIIF, TFIIE, and TFIIH. Finally, Pol II is phosphorylated to proceed to elongation [24]. The transcriptional machinery produces a central portion of RNA during transcription elongation. Transcription ends with the transcription termination process, and RNA travels for further processing and splicing.

Different cell types at constant rates express a limited set of genes called housekeeping genes. However, for cells to differentiate and perform cell-type specific functions, specific genes need to be transcribed at a particular time and

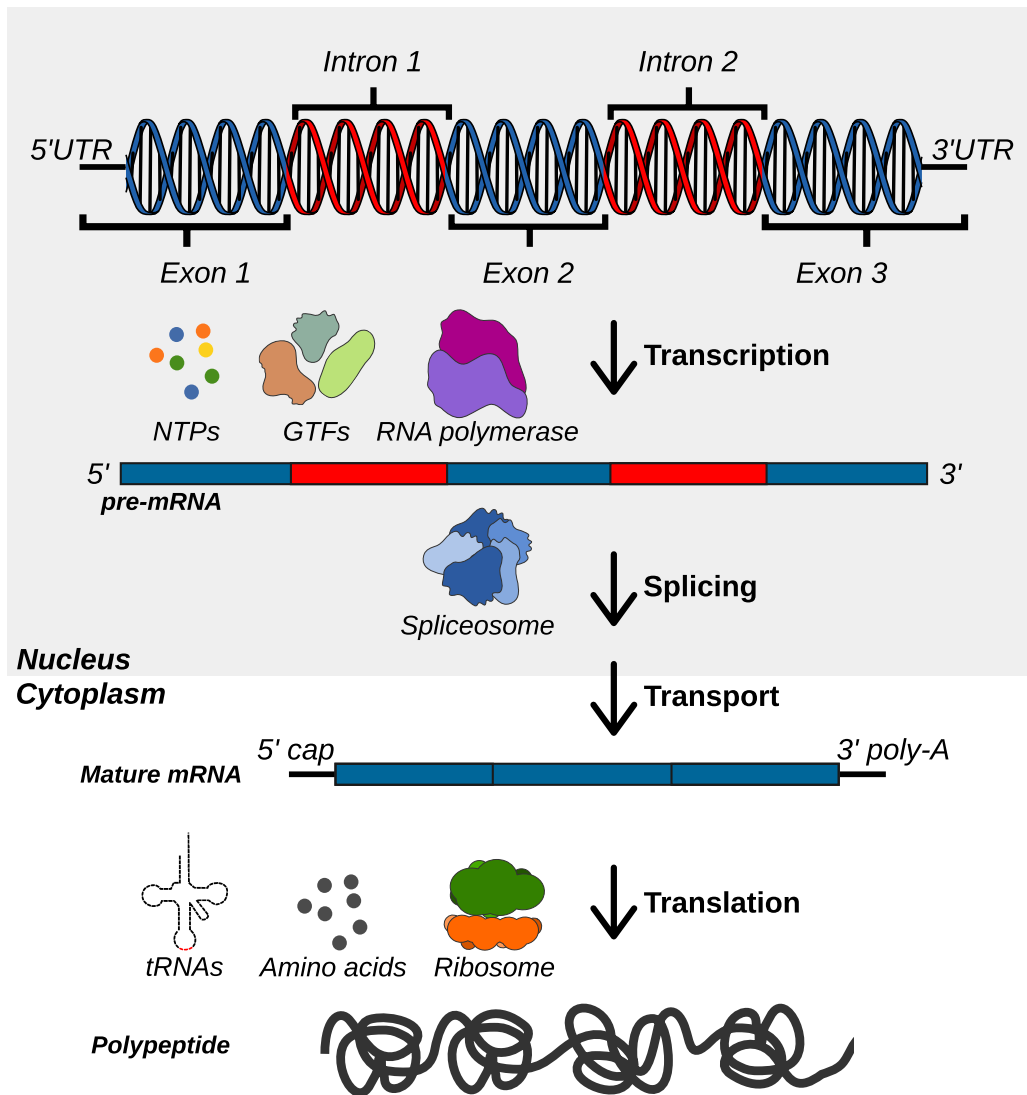


Figure 1.4: From gene to protein. During transcription, an open reading frame (ORF), flanked by 5' and 3' untranslated regions (UTRs), is transcribed by RNA polymerase to produce pre-mRNA. General transcription factors (GTFs) are part of the transcription complex, and RNA polymerase uses nucleotide triphosphates to produce the RNA molecule. Pre-mRNA has introns and exons. The spliceosome cuts out non-protein coding introns during splicing, leaving only protein-coding exons. 5' cap and 3' poly-A tail are added to produce a mature mRNA. In the cytoplasm, the ribosome translates mRNA into a polypeptide chain. tRNAs bring each amino acid into the ribosomes.

Chapter 1. General background

place. Basic transcriptional machinery cannot decide what genes to express, but various regulatory processes make those instructions. For example, TFs bind DNA at the regulatory regions. These regions are promoters (proximal to the TSS), enhancers (distal from the TSS), silencers, and insulators. TF-DNA interactions signal the PIC assembly at the associated gene promoter. Mechanisms of TF binding to the DNA and transcriptional regulation are very complex, and the field still lacks details on the organization of these processes. The next chapter overviews the regulatory regions, TFs, and transcription regulation.

Chapter 2

Transcription factor-DNA interactions

Regulatory proteins and other small molecules are constantly interacting with genomic DNA. The significant players in transcriptional regulation are transcription factors (TFs). Without TFs, RNA polymerase and the basic transcriptional machinery would be unable to approach gene transcription start sites and begin transcription. TFs act in various modes that enable them to locate and bind to regulatory regions: promoters and enhancers. Promoters are genomic regions located near the transcription start site of a gene and serve as the binding sites for RNA polymerase and the transcription machinery. The Mediator protein complex is a significant part of the machinery and facilitates transcription initiation. Enhancers can be located large distances away from the genes they regulate and act to increase or repress the transcription of associated genes. Bound TFs attract Mediator proteins and Pol II, forming a loop that brings enhancer and promoter regions into close spatial proximity, despite potentially being separated by kilobases in linear space (as illustrated in Figure 2.5). This looping mechanism is essential for activating gene expression, which is tightly regulated in a cell-type- and time-specific manner. The cooperation between two or more TFs is necessary to ensure the complexity of transcription regulation. Such interactions are crucial for precisely controlling transcription in response to various signals. This chapter will overview the cis-regulatory regions, transcription factors, and interactions between them.

2.1 Cis-regulatory regions

Regulatory regions in the eukaryotic genome are DNA segments associated with the capability to increase or repress transcription. These regions are termed cis-regulatory regions (CRRs). They are crucial for gene expression regulation because they contain short TF-targeted DNA sequences called transcription factor binding sites (TFBSs) [26]. CRRs include promoters, enhancers, silencers, and insulators.

2.1.1 Promoters

Gene promoters are DNA sequences located near the TSSs of genes (Figure 2.5A). Promoters provide binding sites for RNA polymerase and other transcriptional

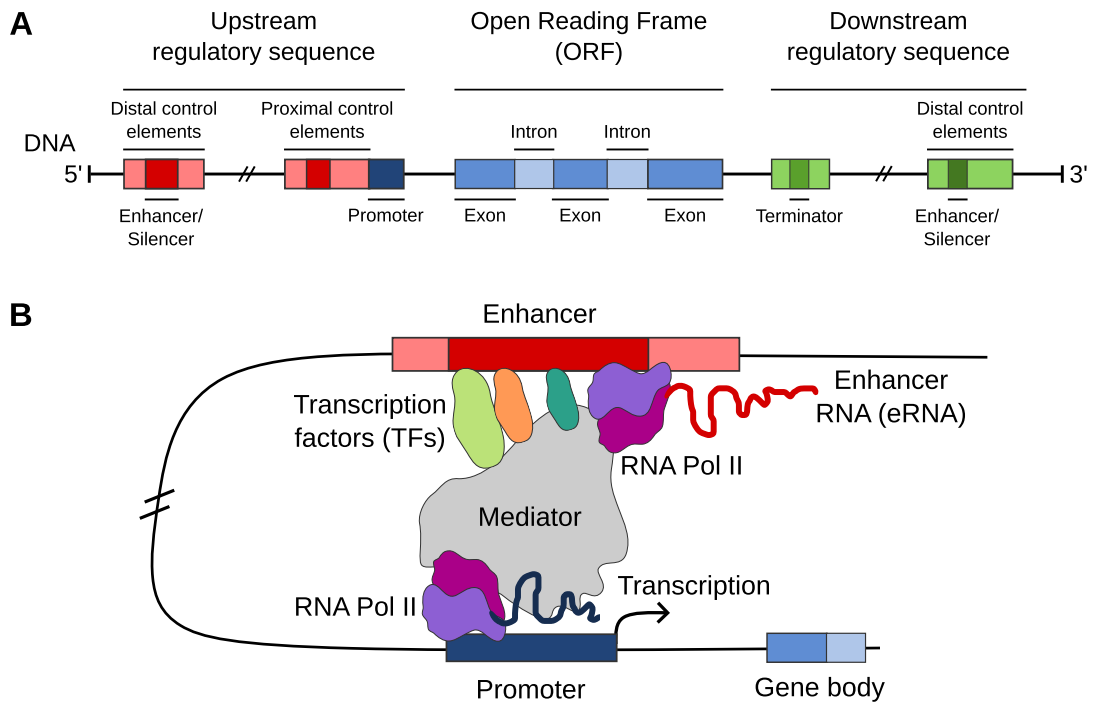


Figure 2.5: Transcription is regulated through protein binding to the gene cis-regulatory regions. **A.** Upstream and downstream of the gene body (open reading frame) lie regulatory regions. Regulatory sequences can be nearby (proximal) or separated by numerous bases (distal). Distal regulatory regions are enhancers and silencers. **B.** For transcriptional machinery to know which gene to transcribe, transcription factors (TFs) are binding enhancers, which stimulate the binding of Mediator proteins and, subsequently, RNA polymerase II (Pol II). The binding brings enhancer and promoter regions closer in space, and transcription starts. RNA Pol II proteins transcribe a gene into pre-mRNA and produce enhancer RNA (eRNA).

machinery proteins to initiate transcription. Core promoters usually extend around 40 bp upstream or downstream of the TSS and are minimal regions needed for transcription initiation [27, 28]. Promoters contain specific DNA sequence elements essential for gene regulation. For example, TBP binds the TATA box element in the core promoter. The Initiator sequence is one of the most common elements and encompasses the TSS [27, 29]. Sequence-specific TFs can bind their sequence motifs in promoters [26–29]. Another characteristic common for human promoters is the abundance of CpG dinucleotides. CpGs are targets for cytosine methylation, one of the gene regulatory mechanisms [26, 30]. Hypermethylation of the CpG dinucleotides in promoters is associated with silent genes, while unmethylated CpGs are found in actively expressed gene promoters [31]. Around 70% of human promoters near TSSs contain a CpG island (CGI), which is a region with a high frequency of CpGs [27, 30, 32].

Based on promoter composition, nucleosome positioning, and what genes they regulate, promoters have four main types: type I, II, III, and IV (Figure 2.6) [27]. Type I promoters are adult tissue-specific, and their transcription starts in a precise position. These promoters have a TATA box around 30 bp upstream from the TSS and have disordered along the sequence nucleosomes. Type II promoters control the expression of the housekeeping genes, and transcription initiation occurs at several locations. Type III promoters regulate developmental genes and also have a broad initiation pattern. These promoters depend on the input from distal enhancers. Type IV promoters are transcriptional machinery-specific and, like type I, have a sharp transcription initiation pattern. Type II, III, and IV promoters have a nucleosome-free region with precisely two nucleosomes positioned downstream and upstream. In addition, all three types of promoters have CGIs covering the TSSs [27].

2.1.2 Enhancers and silencers

Enhancers are distal regulatory regions that activate transcription by acting on a promoter over large genomic distances (Figure 2.5) [33, 34]. They are found upstream or downstream of genes, sometimes located within introns [34]. The distance from the enhancer to the promoter it acts on can vary from tens of bases to megabases [15, 35, 36]. The size of enhancers is typically a few hundred base pairs [37]. Enhancers contain one or more transcription factor binding sites (TFBSs). These are short DNA segments recognized and bound by TFs. The various distribution of TFBSs within enhancers allows TF cooperativity [37]. Combinatorial TF binding enables different TFs to interact and activate or repress the expression of the associated gene. This topic is overviewed in detail in the section about cooperative TF binding.

Enhancers often act spatially and temporally restrictedly, meaning that enhancers are cell-type specific and active only at particular time points [36]. Transcription factors interact with enhancers and bind specific DNA patterns. Multiple TFs can bind one enhancer, depending on how many potential TFBSs that enhancer has [36, 38]. Enhancers produce non-coding RNA, termed enhancer RNA (eRNA) (Figure 2.5B) [33, 39]. These RNAs promote the functions

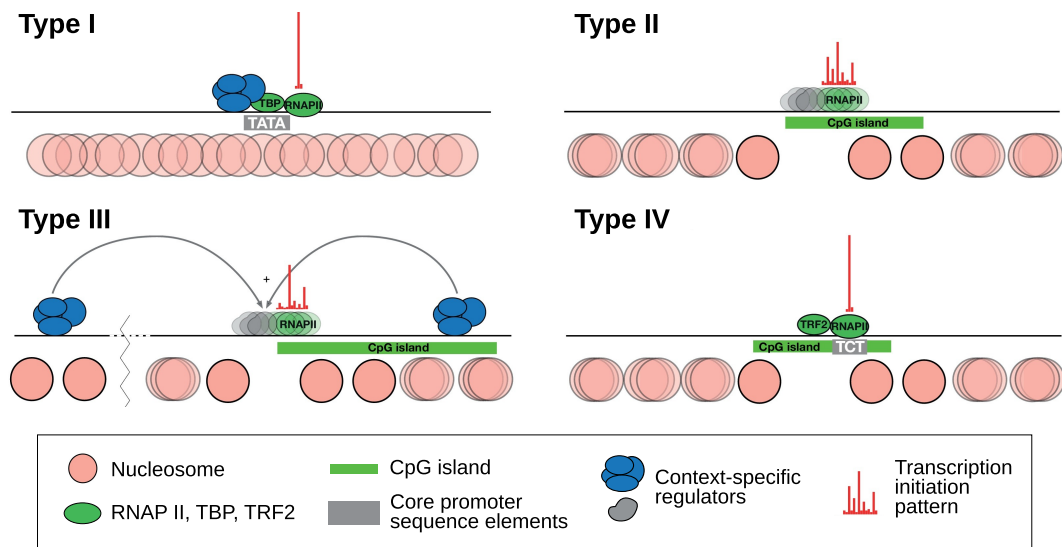


Figure 2.6: Four main classes of promoters. Type I and IV promoters have sharp transcription initiation patterns, and transcription starts in a precise position. Transcription start varies in Type II and III promoters. Type II, III, and IV promoters have two precisely positioned nucleosomes and a CpG island. Type I and II promoters have core promoter sequence elements (TATA box or TCT). Type IV promoters do not depend on the localization of context-specific regulators, while others do. Adapted from [27].

of enhancers by recruiting TFs, promoting enhancer-promoter looping, and influencing chromatin conformation or histone acetylation status [36, 38].

Enhancer-promoter looping occurs in isolated chromatin structures: loops and TADs (Figure 2.7). Cohesin is a ring-like structure that holds the chromatin loops together, letting enhancer and promoter communicate through complexes of proteins [33, 36, 40]. The formation of loops is known as loop extrusion. Another loop-extruding factor is protein CTCF. It binds its specific DNA motifs at the TAD boundaries, this way informing where to stop the loop extrusion process, closing the entire domain (Figure 2.7A) [40]. A single TAD consists of multiple loops held or not by cohesin [10, 15, 40] (Figure 2.7B-C). CTCF binding motif orientation is essential for the loop extrusion since two CTCF proteins bound convergently are required to close the loop [40]. The process insulates a particular region [36]. The influence of the enhancers is mostly, but not always, limited to the genes within the same loop [10, 15, 41]. One enhancer can regulate multiple genes, while multiple enhancers control one gene.

Opposite to the enhancers, the repressor proteins bind silencers to suppress the transcription [37]. Silencers can also be tissue-specific and operate within TADs through short- or long-range interactions [42, 43]. These regions are marked by specific chromatin marks, for example, Histone 4 Lysine 20 methylation (H4K20me) [42, 44]. Repressive TFs target silencers and, through looping, communicate with the gene's promoter [43].

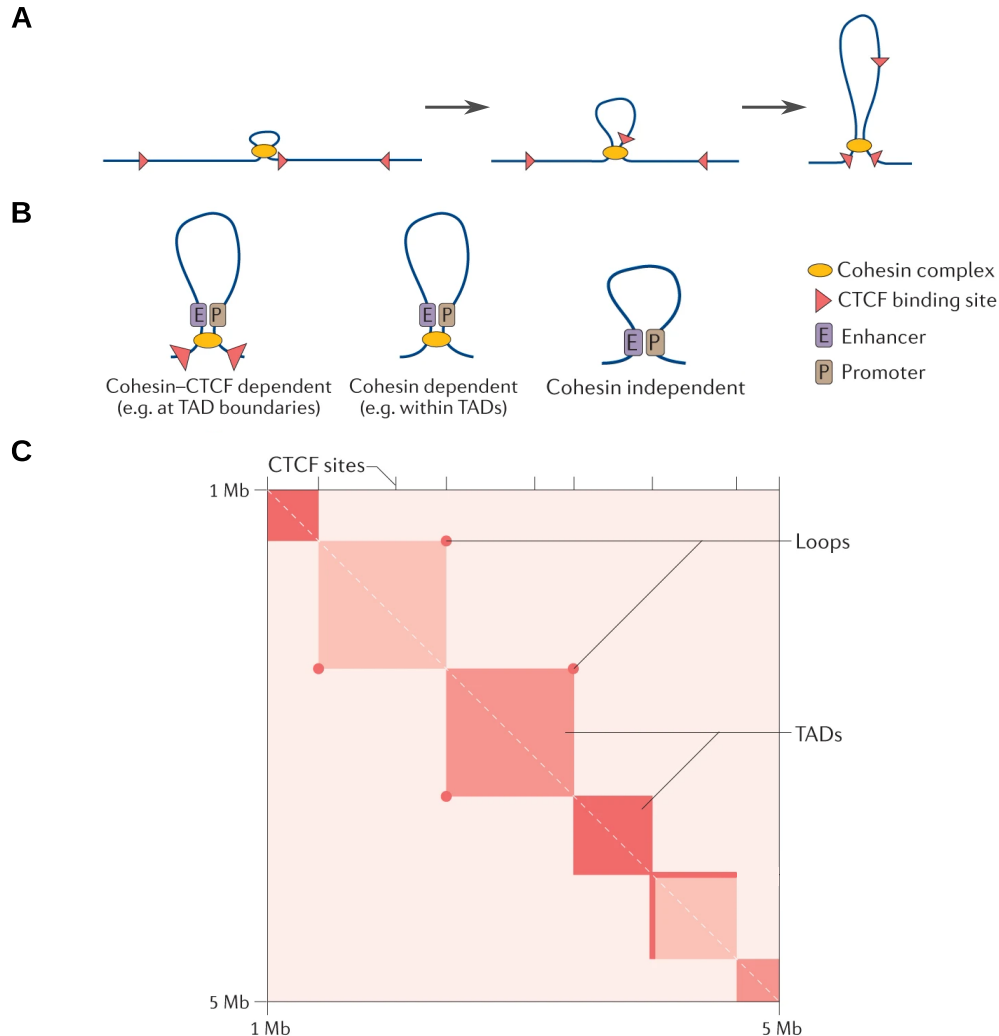


Figure 2.7: Chromatin is forming loops and topologically associated domains (TADs) using looping factors, such as cohesin and CTCF binding, to enable and spatially control promoter and enhancer communication. A. During loop extrusion, cohesin associates with DNA, and the loop is symmetrically extruded until cohesin encounters convergently oriented CTCF molecules. **B.** Loops can be cohesin and CTCF-dependent, marking the TAD boundaries (left). Within TADs, loops can be only cohesin-dependent (center) or entirely independent (right). **C.** A schematic heatmap represents the DNA-DNA contacts in a 4 Mb region in the mammalian chromosome. The more intense the color, the more frequent the contacts. TADs organize the contacts, and loops mark the TAD boundaries. Adapted from [40].

2.2 Transcription factors

Transcription factors (TFs) are key transcription regulators [45, 46]. Around 1,600 human genes are encoding for TFs [47, 48]. TFs act as a bridge between the cell signaling pathways and gene expression. Through receptors, external cues trigger post-translational modification cascades that eventually result in the modulation of a set of TFs, which are then recruited at the gene cis-regulatory regions [46]. Depending on the stimuli, cell type, and state, TFs are modified. Short DNA sequences recognized by TFs are known as transcription factor binding sites (TFBSs) [47]. TF DNA sequence specificities and preferences are summarized as motifs. The function of TFs and their recognized DNA motifs are often studied together. Various experimental methods exist to identify TF-DNA interactions and various computational tools to model and represent those interactions. Most of the experimental techniques are limited to investigating individual TFs. However, in reality, TFs often cooperate, compete for the same binding site, or interact with the DNA indirectly [45]. These complex TF activity mechanisms are still challenging to investigate and interpret with currently available tools.

2.2.1 Structural classification

Eukaryotic transcription factor proteins have at least two domains: DNA binding and activation or repression. Optionally, there are protein-protein interaction (PPI) or effector domains to regulate TF activity [49–51]. DNA-binding domain (DBD) recognize and bind specific DNA patterns (motifs). Many protein-DNA complex structures are available in resources, such as Protein Data Bank (PDB) [51–53]. For example, the basic helix-loop-helix (bHLH) domain proteins have two α -helices connected by a turn. These domains are known to dimerize, and each monomer recognizes one side of a symmetric DNA sequence (Figure 2.8A). bHLH proteins bind at the major groove of the DNA molecule [52]. Another example of a DBS is homeodomain-type domains. POU5F1, also known as OCT4, has two homeodomain DBDs, which are binding cooperatively at the major groove (Figure 2.8B).

DBDs are extremely well-conserved and are used to classify TFs into families and classes. TF class more broadly describes TFs based on the DBDs, while the TF family categorizes TFs further based on their sequence similarity and evolution. The similarity of amino acid sequences forming a DBD is the basis of the TF classification. TFs from the same class or family tend to recognize the same or highly similar DNA motifs (Figure 2.8C) [45]. It is an important feature that can help to define binding specificities of understudied TFs (a process called DNA binding motif inference) [51]. In addition, DBD can provide information about the protein function. For example, different homeodomain DNA binding domain-containing TFs often regulate developmental processes [54].

In 2009, a group of researchers defined and described 23 human TF families, where 80% of all known human TFs at the time had C₂H₂ zinc-finger, homeodomain, or HLH DBDs [54]. Later classifications expanded from humans to all mammalian TFs. One of the most extensive classifications of mammalian

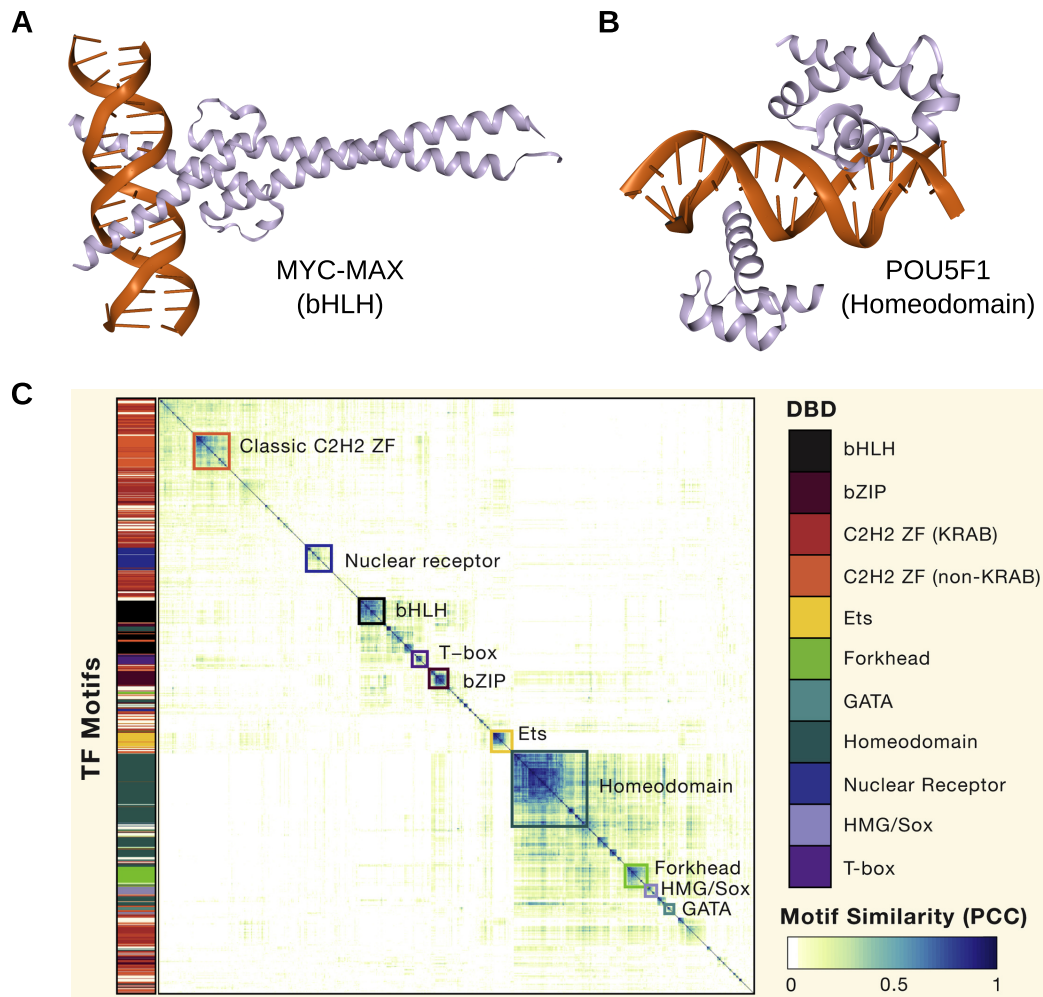


Figure 2.8: Transcription factors with the same DNA binding domains (DBDs) bind similar DNA sequence patterns. **A.** 3D model of *Homo sapiens* TF dimer MYC and MAX DBD binding to the DNA (PDB ID: 1NKP). MYC and MAX belong to a basic helix-loop-helix (bHLH) TF class. Proteins in purple, DNA in orange. **B.** Same as A, but *Homo sapiens* POU5F1 DBD interaction with DNA (PDB ID: 8G87). POU5F1 belongs to a homeodomain TF class. **C.** Often, TFs with similar DBDs are binding to highly similar DNA motifs. A heatmap represents human TF motif pairwise comparisons. Motifs group according to the protein DBDs. Adapted from [47].

transcription factors, TFClass, includes superclass, class, family, subfamily, and genus categories [48, 51, 55, 56]. For example, in the JASPAR database (storing TF-DNA binding computational models), TF families and class categories are the most used to annotate TF-DNA binding profiles. The most up-to-date version classifies more than 39,000 TFs from up to 41 mammalian species. The TFClass annotates TF DBDs and classifies TFs using phylogenetic analyses [55]. More recently, a similar classification of plant TFs was released. Many DBDs are present in plants but not mammals, which hindered the complete classification of the plant TFs. Plant-TFClass addressed that by adding eight new classes and 37 new families to classify most of the 56 plant TF types [57]. This recent development in TF classification highlights the need to consistently revise and update current classifications, also considering other taxa, such as nematodes or insects.

2.2.2 Functional classification

TFs are also classified based on their functional roles. The main groups are pioneer, settler, and migrant TFs [58]. In brief, pioneers can interact with the nucleosome-protected closed chromatin and trigger its opening for other TFs and proteins [58–60]. Settlers then bind the accessible adjacent DNA established by the pioneer TFs and maintain the gene expression [58, 61]. On the other hand, the migrants can quickly bind and unbind the DNA in response to the cellular signals [58]. These three groups coordinate the precise gene expression together.

Pioneer TFs are also known as master regulators [60]. Mouse HNF3 (FoxA) and GATA-4 were the first factors detected to engage the enhancers in development, and their occupancy preceded occupancy of the other TFs that bind to those enhancers [62–64]. Well-characterized pioneer TFs include SOX2, KLF4, and OCT4, also known as Yamanaka factors, which can induce pluripotency [64]. Pioneers bind the condensed chromatin before the transcription activation, therefore showing their essentiality for region transcriptional activation [64]. In some cases (p53) (SOX2, KLF4, and OCT4), it is enough for a pioneer TF to recognize a partial DNA motif to bind closed chromatin; in other scenarios, pioneers prefer binding near the nucleosome edge, and their binding affinity is more substantial than to the sites located in the naked DNA [64]. After binding, pioneer TFs open closed chromatin locally first. Then ATP-dependent chromatin remodelers, such as the SWI/SNF remodeling complex, are recruited to promote nucleosome eviction in the region [59]. Finally, acetyltransferases modify histone tails as a product of a regulatory region activation process [59, 65]. In cancers, some pioneer TFs promote DNA region demethylation [66]. Aberrant methylation modulates chromatin structure and transcription of cancer-related genes.

Settler transcription factors are pivotal in maintaining and stabilizing gene expression patterns established by pioneer transcription factors. They depend on the open chromatin at their DNA binding sites for binding and almost always bind to genomic sites matching their DNA binding motif [58, 61]. Examples of settlers include Myc/MAX, nuclear hormone receptor, AP-2, and NF κ B TF families [61]. Settlers and pioneers bind hierarchically. The loss of pioneer TF binding results in chromatin closing and, therefore, loss of settler TF binding [61].

Finally, a majority of all transcription factors are migrants. They bind only sporadically to a subset of their target sites, even in the open chromatin [58, 61]. Furthermore, migrants depend more on specific co-factor interactions [61].

2.3 Transcription factor binding to the DNA

Transcription factors bind cis-regulatory regions (promoters and enhancers) to regulate associated gene transcription. TFs typically recognize short (10-20 bp) DNA sequence motifs [67]. Depending on the sequence context and available co-factors, TFs can either activate or repress transcription of a gene [47]. Interactions with CRRs are highly cell-type-specific. The location where a TF binds DNA is called a transcription factor binding site (TFBS). A promoter or an enhancer can have multiple TFBSs of the same or different TF. Transcription factors generally have a binding preference for particular DNA patterns, also known as DNA motifs [65].

How do transcription factors find these small binding sites in a long DNA molecule surrounded by various proteins and molecules in the nucleus? The four primary forms of TF motion in the environment with DNA are (1) 3D Brownian diffusion in the nucleoplasm, (2) 1D sliding along the DNA, where TF binds DNA in a non-specific fashion, (3) already bound TF binds a nearby region (intersegmental transfer), (4) TF unbinds from a region, briefly diffuses, and rebinds to a nearby section of the same region (hopping) [68]. Each motion takes a different time for a TF to find a target region.

The way a TF interacts with the DNA influences transcription regulatory mechanisms. TF binding varies from direct binding to the DNA (most well-studied) to indirect and cooperative binding [65, 69]. TF binding to the DNA depends on various features that co-exist together. For example, DNA modifications, such as DNA methylation, can dictate whether a TF will bind. DNA shape, presence of co-factors, or sequence variants can also determine TF binding [69]. It is essential to understand these mechanisms. The different binding modes can increase the complexity of gene transcription regulation. It is necessary to ensure cell-type-specific transcription.

This section will overview direct, indirect, and cooperative TF binding, the impact of pioneer TFs, and DNA methylation. Various experimental techniques enabled the deciphering of TF binding mechanisms, and the section will briefly describe the most popular and widely used methodologies.

2.3.1 Direct and indirect TF binding

TFs directly bind DNA when they interact with the DNA in a sequence-specific manner. In such cases, the protein recognizes a specific DNA motif. However, the affinity of this interaction can vary. In other instances, TFs can bind the DNA indirectly. It primarily describes the binding through the different proteins via protein-protein interactions (PPIs) [70]. Some TFs can bind both directly and indirectly. For example, transcription factor Spi-1/PU.1 can bind to DNA directly and indirectly. Direct binding of Spi-1/PU.1 to the consensus motif *AGGAA* leads

to transcriptional activation, while indirect binding through a partner GATA1 to the *GATA* motif may lead to transcriptional repression [65].

Experimental techniques that map TF-DNA interactions *in vivo* capture direct and indirect TF binding. Therefore, careful analysis of those regions and interpretation of the results are required. DNA motif analysis is the simplest way to determine whether the TF is binding directly (canonical motif present) or indirectly (lack of the canonical motif) [70]. A section in the next chapter discusses computational approaches to distinguish direct from indirect TF-DNA interactions.

2.3.2 Cooperative transcription factor binding

Cooperativity of TFs on regulatory regions, such as enhancers, is an essential transcription regulation mechanism [60]. Interaction with other sequence-specific or non-DNA binding co-factors can alter binding specificity [69]. Cooperative binding does not always mean that two TFs will bind the DNA directly; they can cooperate through protein-protein interactions or bind DNA indirectly [45].

We can analyze the cooperativity between TFs with DNA binding specificity through their binding DNA patterns. The rules to describe the cooperativity between the TFs are often referred to as motif grammar [60]. It is the presence of specific motifs, their order, the spacing between them, and the orientation of those motifs (Figure 2.9) [60]. Particular order, motif orientation, and spacing enable direct protein-protein interactions, promoting cooperation. The two motifs can also overlap, thus creating a composite motif co-bound by a dimeric TF complex. Such cases are often challenging to investigate due to the limitations of experimental and computational tools. Some TFs require a rigorous motif grammar for their cooperativity. Others bind with various spacings, described as a relaxed grammar or soft syntax [71]. The orientation of the DNA motifs alone can be a strong determinant of transcription activity. For example, Spi-1/PU.1 modulates gene expression in erythroid cells and promotes oncogenesis. Research has also shown that it cooperates with KLF family TFs. When the Klf-1 motif is in direct orientation in the promoter regions, Spi-1/PU.1 activates transcription, while the genes are repressed when Klf-1 binds the motif in reverse orientation [65].

Currently, three main models describe cooperative TF binding at the enhancer regions: enhanceosome, billboard, and TF collective model (Figure 2.10) [37, 60, 72]. The two main models are enhanceosome, which requires a rigid motif grammar, and the billboard model, which allows soft motif syntax and the presence of specific TFs is more important than the order, orientation, and spacing of the motifs (Figure 2.10A-B). Regulatory regions bound by TFs using an enhanceosome model are highly conserved in evolution, and TFs bind directly to strictly organized motifs. (Figure 2.10A). Precise binding site orientation, order, and positioning are required for an enhancer to function [72]. Regions following the billboard model are less conserved, and cooperativity is indirect, but TFs also bind directly to the DNA motifs (Figure 2.10B). The positioning, orientation, and distance of cognate motifs and their order vary without impacting the activity of the enhancer. The

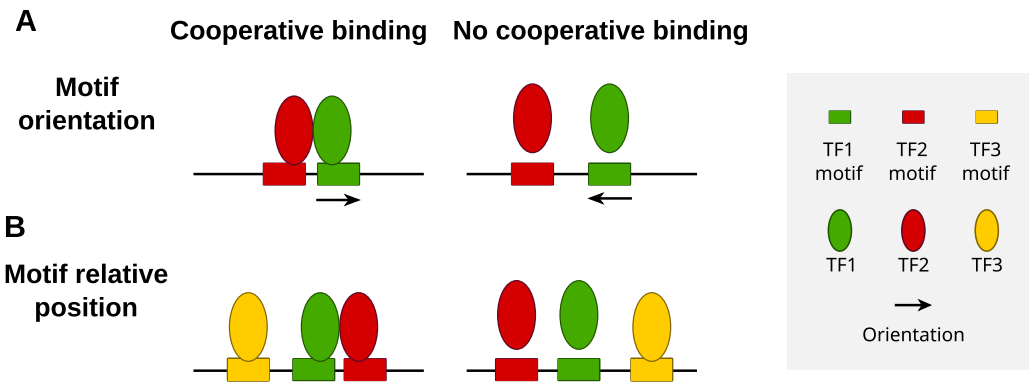


Figure 2.9: Motif grammar facilitates transcription factor cooperativity. TF cooperativity is determined by **A.** motif orientation and **B.** motif relative position and spacing. **A.** When TF1 binds a motif in a preferred orientation (left), it interacts with TF2 directly, promoting cooperative binding. No cooperative binding occurs when the motif has a different orientation (right). **B.** Direct protein-protein interaction can occur when three TF motifs are spaced in a specific order (left), while PPI cannot occur when the spacings and order change (right). Adapted from [37].

presence of the TF binding sites is the most important aspect [72]. The third model, TF collective, falls somewhere in between the first two (Figure 2.10C). In this case, the regions are also less conserved. TFs can bind directly and indirectly; motifs are not always present, and they have flexible organization. Finally, one enhancer can include a mixture of all models. Such regions have a variable sequence conservation, TFs can bind directly and indirectly, some motifs have a strict grammar, some have more flexible organization, or motifs are not present at all [60].

Understanding how different transcription factors work together to regulate transcription is essential. The cooperativity of TFs provides an incredible complexity of regulatory mechanisms, which allows spatial and temporal control of cell differentiation and growth. Specific experimental techniques, such as CAP-SELEX, have been developed to profile dimeric TF binding (see section experimental methods for TF-DNA interaction detection) but are still limited. Computational tools provide more opportunities to look into TF cooperativity (read more about computational models in the next chapter).

2.3.3 Impact of methylation to the DNA binding

DNA modifications impact how transcription factors bind [73]. The most extensively studied epigenetic DNA modification is DNA methylation. “Writers”, DNA methyltransferases (DNMTs), modify the cytosines in the DNA molecule by adding a methyl group in the context of CpG dinucleotides, generating 5-methylcytosine (5mC) (Figure 2.11A). In contrast, ten eleven translocation (TET) enzymes conduct the removal of methyl groups [73, 74]. Generally, hypermethylation of CpGs in the promoter regions is associated with transcription

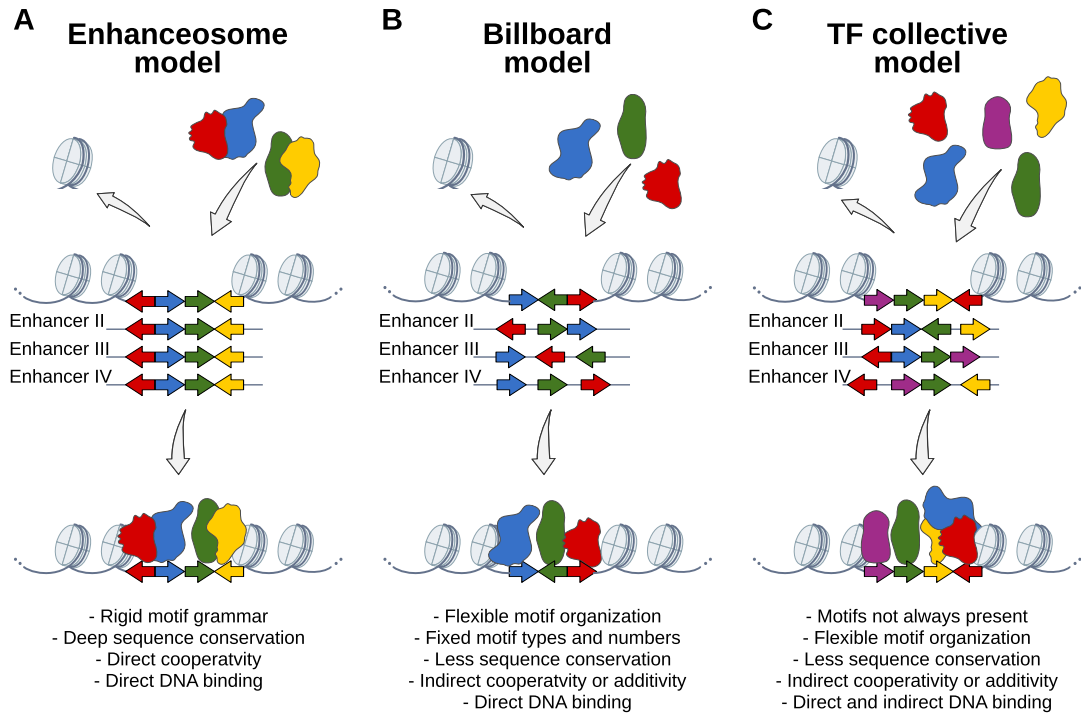


Figure 2.10: Enhancer models for TF cooperative binding. **A.** Cis-regulatory regions that follow the enhanceosome model have rigid motif grammar and deep sequence conservation and are bound directly by directly cooperating TFs. **B.** Regions that follow the billboard model have a much more flexible motif organization but fixed motif types and numbers. These regions are less conserved but also directly bound. In contrast to the enhanceosome model, TFs are cooperating indirectly or additively. **C.** TF collective model has features from both enhanceosome and billboard models. Sequences are less conserved, and motif organization is flexible. DNA motifs are not always present, and TFs can directly and indirectly interact with the DNA. Adapted from [60].

repression, while gene body hypermethylation indicates active genes [74, 75]. Therefore, DNA methylation is essential for TFs that bind methylated motifs, and some prefer to interact with methylated sites [74, 76]. For example, the HOXB13 protein has three key residues that contribute to the preferential mCpG binding (Figure 2.11B-C) [76]. HOXB13 can also bind to the unmethylated motif, which then does not contain a CpG dinucleotide (Figure 2.11C). Some TFs (for example, MAX) are not influenced by methylation even though their motifs contain a CpG [76]. For other TFs, DNA hypermethylation suppresses their binding. Changes in the methylation patterns can lead to aberrant TF binding, affecting transcription regulation and possibly causing diseases.

Most common TF-DNA interactions computational models do not consider methylated CpGs and position interdependency. The dependency between the positions in the TF-recognized DNA sequence is of particular relevance when analyzing CpG dinucleotides. A CpG in the binding motif does not inform whether it needs to be methylated for the TF to bind. There are specific attempts to model TF binding to the methylated DNA. For example, the MEDEMO toolkit analyzes TF binding motifs by including information about DNA methylation [77]. The simplest way to include methylation information is to extend the alphabet where 5mC is replaced with M and G on the opposite strand with an H [77, 78]. Methylation-aware TF binding models can use such methylation-aware genomes for training. These can, therefore, inform about methylation sensitivity (Figure 2.11D) and allow the identification of differential TF binding regions in ChIP-seq data [77]. The modeling can also include other DNA modifications (5-hydroxymethylcytosine, 5-formylmethylcytosine, or 5-carboxymethylcytosine) by coding them as specific letters [78].

2.3.4 Experimental methods for TF-DNA interaction detection

Over the years, scientists developed multiple experimental methods to detect the binding preferences of transcription factors. Broadly, these methods are applied *in vitro* or *in vivo*. Most of the techniques, though, are limited to investigating the interactions of a single TF at the time. This section will describe the most popular and relevant methods. In addition, it will explain a few techniques that probe open chromatin. Information on the open chromatin can also define the transcription factor binding landscape.

In vitro

In vitro methods mainly aim to identify TF binding sequence preferences. For example, systematic evolution of ligands by exponential enrichment (SELEX) or protein binding microarray (PBM). Other methods, such as EMSA, can inform about the binding affinity [79]. The most significant limitation of *in vitro* methods is that they cannot give information about the biological context of the TF-DNA interaction by design.

SELEX is a qualitative method developed decades ago, but high-throughput (HT-SELEX) and other versions are extensively used today [80, 81]. The basic principle of SELEX is the incubation of a purified TF with a pool of random

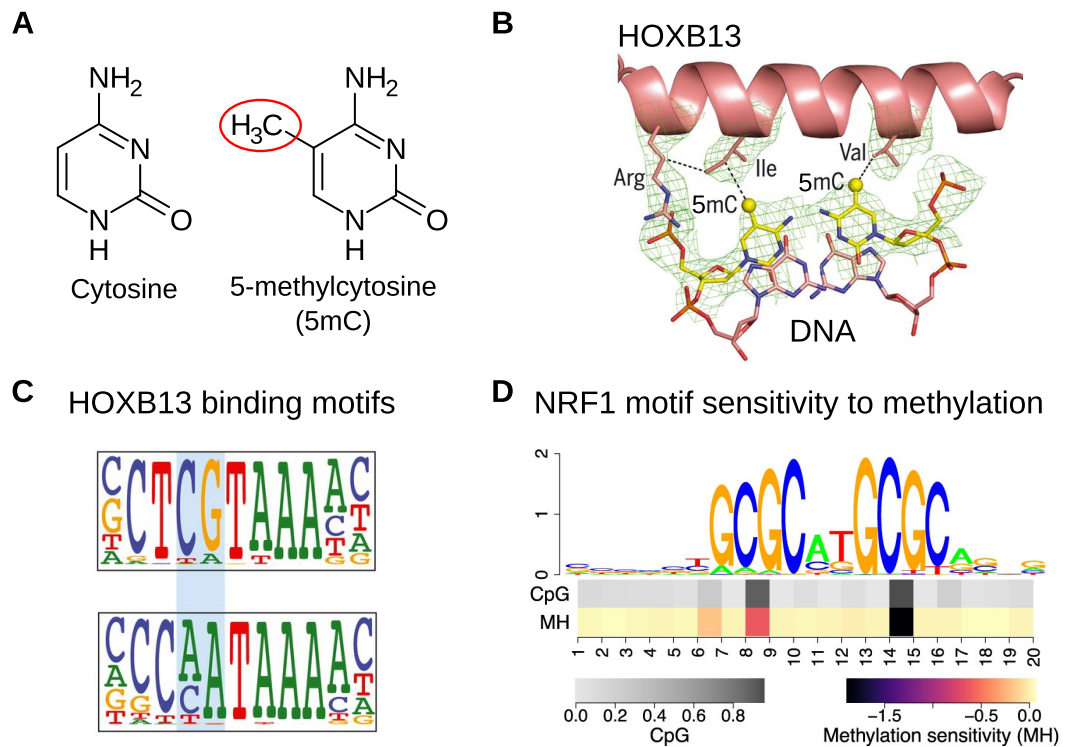


Figure 2.11: TF binding is affected by DNA methylation. **A.** DNA methyltransferases add methyl- group to the cytosines in the CpG context and generate 5-methylcytosine (5mC). **B.** Some TFs, such as HOXB13, specifically recognize DNA sequences with methylated cytosines. Three key protein residues, Arg, Ile, and Val, direct the specificity towards 5mC. **C.** HOXB13 prefers binding to the motif containing a methylated CpG (primary motif; above) but also binds a motif without a CpG with lower affinity (secondary motif; below). **D.** Some specific methodologies investigate TF binding sensitivity to methylated sites. For example, NRF1 is sensitive to methylation and binds to a motif with CpG content. Based on the sites bound by NRF1, NRF1 does not prefer methylated CpGs. The first CpG is less affected by methylation than the second. Adapted from [76, 77].

DNA oligonucleotides (Figure 2.12A). Random sequences bound to the protein are collected, amplified, and incubated with the TF again. After multiple cycles of incubation, separation, elution, and amplification, the sequences with the highest affinity are selected and sent to the sequencing procedure. The resulting sequencing data is analyzed to determine the TF binding sequence preference [79, 81]. Differently from a regular SELEX, one can study multiple transcription factors simultaneously in HT-SELEX. However, since the oligonucleotides are random, the sequences bound with the highest affinity in SELEX (or HT-SELEX) are not guaranteed to reflect the functional binding preferences in a living cell.

One variation of the SELEX protocol is particularly relevant because it allows for studying dimer TF sequence preferences. In the consecutive affinity-purification systematic evolution of ligands by exponential enrichment (CAP-SELEX) protocol, oligonucleotides are incubated with two TFs, flagged with different flags. During two consecutive affinity purification steps, only those sequences bound by both TFs proceed to amplification [82]. This technique shows how specifically two TFs can bind the same sequence. For example, we can determine preferred spacings or overlaps between two individual motifs and motif orientation [82].

PBM is a high-throughput method where synthetic DNA oligomers are enriched with a tagged TF, like in SELEX, to characterize TF-DNA interactions comprehensively [83]. The three main PBM experiment stages include primer annealing, primer extension, and protein binding. DNA microarrays immobilize numerous short DNA sequences containing many potential TF binding sites. Assayed DNA binding protein is expressed with an epitope tag and then purified. Protein directly binds to triplicate double-stranded DNA microarrays [83]. The specific binding is quantified by fluorescence intensity. Analysis of the binding indicates the DNA binding site sequence specificities of TFs *in vitro* [83].

A high-throughput technique falling in between *in vitro* and *in vivo* is called DNA affinity purification sequencing (DAP-seq) [84]. DAP-seq uses native DNA to construct the libraries of sequences, preserving cell-specific modifications. Native DNA libraries provide increased specificity compared to methods such as SELEX. The library of sequences is then incubated with *in vitro*-expressed and tagged TF. TF-bound sequences are purified and sequenced. Finally, sequences are mapped to the origin genome to identify locations preferentially bound by a TF [85]. DAP-seq is an advanced version of the ampDAP-seq (amplified DAP-seq), which uses a PCR-amplified genomic DNA library for protein binding [84, 85]. The amplification step enhances sensitivity, allowing the detection of weak or transient TF-DNA interactions. However, amplification also removes all secondary DNA modifications, such as DNA methylation, potentially preventing modification-sensitive TF-DNA interactions. Therefore, DAP-seq is better for investigating TF binding to the modified DNA.

In vivo

The most popular technique to map TF-DNA interactions in living cells is chromatin immunoprecipitation followed by sequencing (ChIP-seq) [86]. The ChIP-seq process starts with the cross-linking of the DNA and proteins in the living cells using formaldehyde (Figure 2.12B). Then, DNA is randomly fragmented with

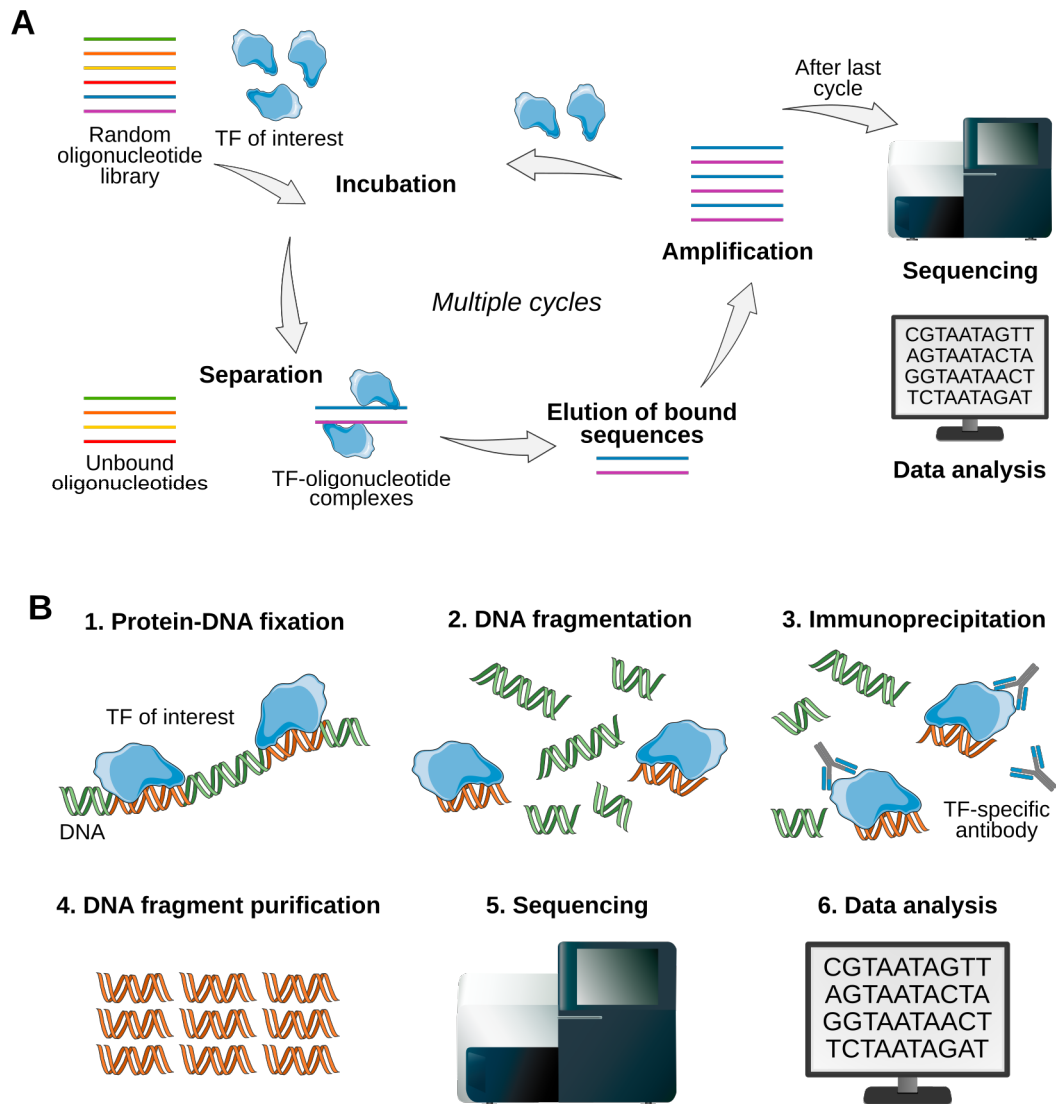


Figure 2.12: Examples of experimental techniques to investigate TF-DNA interactions. **A.** SELEX is an *in vitro* method that allows the determination of the TF binding sequence preferences. In the procedure, a pool of random oligonucleotides is incubated with the TF of interest. Unbound oligonucleotides are separated from TF-DNA sequence complexes and washed away. Bound sequences are purified, amplified, and incubated again. The incubation cycle is repeated several times, and after the last cycle, sequences bound with the highest affinity are sequences, and the resulting data is analyzed. **B.** ChIP-seq is an *in vivo* technique for identifying genomic locations bound by a TF of interest. DNA and proteins are cross-linked (1), and DNA is fragmented (2). The protein of interest is immunoprecipitated using a specific antibody (3). These DNA-protein complexes are selected, and DNA is purified (4). Eluted DNA is sequenced (5), and the resulting data is analyzed (6).

sonication, leaving DNA segments bound and unbound by proteins. In a crucial step, a specific antibody binds the TF of interest. The washing removes everything that is not TF-DNA complexes bound by antibodies. Further steps reverse protein-DNA cross-links and purify DNA fragments previously protected by the TF. These DNA sequences proceed to sequencing. After sequencing, sequencing reads are mapped to a genome. Further analysis identifies those genomic regions enriched by TF binding [86, 87].

The most significant advantage of *in vivo* techniques, such as ChIP-seq, is that the TF binding locations are determined in a specific tissue or cell line, thus reflecting the TF binding preferences in physiological conditions [87]. However, this comes with a few limitations. TF binding and regulatory mechanisms are complex, so *in vivo* binding is influenced by several factors that affect the outcome of the experiment. Differences in TF-DNA affinity, protein-protein interactions, indirect binding, and sequence biases can all affect how results are interpreted [81, 88, 89]. Other limitations of the ChIP-seq technology include studying only one TF at a time, and the immunoprecipitation requires a high-quality antibody, which can be difficult to acquire if the protein of interest is relatively unknown or poorly annotated [87].

In ChIP-seq, the DNA fragment sizes are variable and are longer (hundreds of base pairs) than the actual TF-protected region because DNA shearing is random [88, 90]. Therefore, the TF binding site inference is not in base pair resolution. Computational analysis and motif search cannot identify all binding locations in ChIP-seq derived regions [88, 91]. Researchers developed the ChIP-exo technique to overcome this limitation. The protocol has an additional step of immunoprecipitate treatment by 5' to 3' lambda exonuclease, which digests the unprotected ends of the DNA fragment bound by a TF. It allows us to get precise, nearly single-base resolution genomic locations bound by a TF [88, 91]. The length of such regions is 36-100 bp [90]. A simplified ChIP-exo protocol, ChIP-nexus, was developed in 2015 [89, 90].

The latest methods for TF binding region identification *in vivo* include CUT&Tag and CUT&RUN [92]. Both techniques are similar and are based on specific antibodies recognizing the TF of interest. In CUT&Tag, the antibody tethers the A-Tn5 transposase fusion protein. After its activation, transposase adds adapters for paired-end DNA sequencing, and high-resolution fragment libraries are generated [92, 93]. In CUT&RUN protocol, micrococcal nuclease (MNase) is tethered by the antibody and, when activated, cuts the DNA at the borders of TF binding [94]. These novel methods allow for small numbers of cells as input and profiling of TFs in single-cells (CUT&Tag is better suited for this application) and promise a more accurate determination of the TF-bound genomic locations by providing high resolution and exceptionally low background [92–94].

Even with new technologies emerging every year, ChIP-seq remains an essential method for TF binding profiling. Many ChIP-seq data are publicly available, but researchers have not fully explored it. It motivates researchers to work on novel computational methods that allow them to study this data further.

Profiling chromatin accessibility

A separate category of experimental techniques aims to investigate open chromatin. Instead of profiling a single transcription factor, techniques such as assay for transposase-accessible chromatin using sequencing (ATAC-seq) or DNase I hypersensitivity (DHS) provide information on regions accessible to proteins, such as TFs.

In the DHS assay, a DNase I enzyme selectively digests nucleosome-depleted bulk DNA [95]. After digestion, DNA fragments are sequenced, and then reads are mapped to the genome. The computational analysis identifies DHS footprints that indicate the nucleosome-free regions, potentially bound by various TFs [95, 96].

ATAC-seq uses a Tn5 transposase to cut and insert sequencing adapters into regions unprotected by nucleosomes or other proteins [97]. The resulting DNA fragments are then sequenced. Sequencing reads are mapped to the genome. From reads, open regions (peaks) are called and analyzed computationally. Computational tools predict the binding of different TFs (more on TF binding site predictions in the next chapter) [97]. The prediction of a precise binding location of a TF at a particular locus is defined as TF footprinting. When a TF binds its binding site, it leaves a "footprint" - the precise binding location is protected by the TF, while the immediate surroundings are affected by Tn5 transposase. Cleavage patterns can be compared between bound and unbound DNA. The data can be analyzed to determine what locations were protected, thus bound, by a TF. Footprinting aids the identification of regulatory regions, such as promoters and enhancers. ATAC-seq can be in bulk or single-cell form. Investigation of open chromatin in single cells is used to define the cell-type specific accessibility, where TFs are accessing different regions [97].

A more recent technique, single-molecule footprinting (SMF), allows probing chromatin accessibility and co-occupancy of transcription factors at single-molecule resolution [98, 99]. The method utilizes cytosine methyltransferases and bisulfite sequencing to resolve the occupancy of Pol II and multiple TFs simultaneously on single DNA molecules. DNA is subjected to methylation in CpG or GpC contexts by using recombinant methyltransferases. They methylate only those cytosines not protected by the TFs, polymerase, or nucleosomes. Bisulfite libraries can be whole-genome or only target regions of interest [99]. Sequencing results provide information on methylation status for each molecule/read [98, 99]. The methylation status can be summarized for each region of interest, for example, a region containing the potential binding sites for two TFs: TF1 and TF2. The more methylated, on average, the genomic region is, the more accessible it is (Figure 2.13A) [98, 99]. Further computational analysis infers five occupancy states for each molecule and region: "TF1 + TF2", "TF1", "TF2", "Accessible", and "Nucleosome". Bins, overlapping and neighboring regions of interest, summarize the methylation of individual CpGs. This binarized data represents different occupancy states (Figure 2.13B).

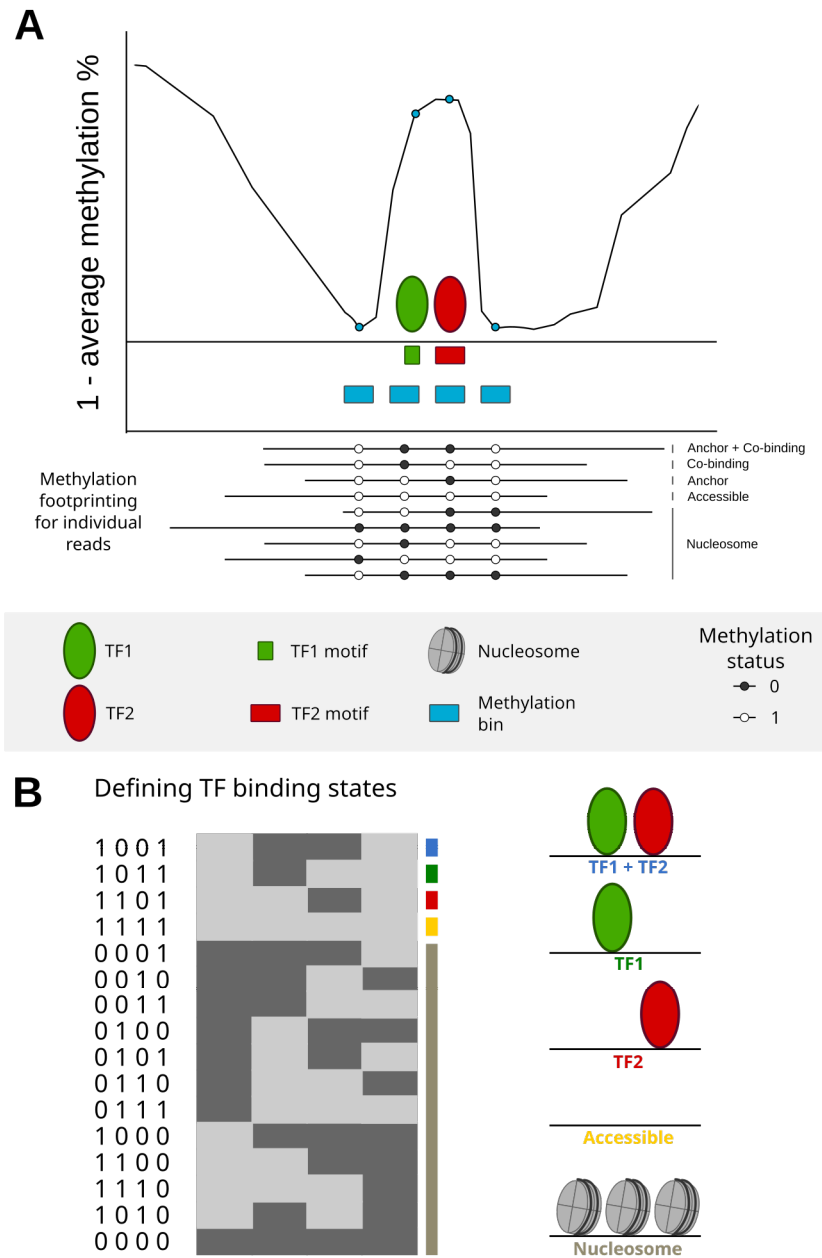


Figure 2.13: Single-molecule footprinting (SMF) technique enables investigation of cooperative events on single-molecules. **A.** SMF results in methylation information for single molecules. Methylation can then be analyzed and summarized for individual regions of interest (for example, the potential binding site for anchor and co-binding TFs, red and green blocks, respectively). The average methylation signal visualized as a line plot gives a general overview of the accessibility, where the lack of methylation represents open chromatin. From SMF data, the occupancy states can be inferred by summarizing methylation in four bins that overlap with regions of interest and neighbor them (blue blocks). **B.** From binarized data, five states for each molecule can be inferred: “TF1 + TF2”, “TF1”, “TF2”, “Accessible”, and “Nucleosome”. Adapted from [100].

Summary of papers

Chapter 4

Summary of papers

This chapter summarizes the three publications that are the foundation of this thesis. The unifying theme of this thesis is the investigation of transcription factor-DNA interactions. I explored this theme through two main aims:

(1) discover new TF binding profiles and update the JASPAR database while also developing and updating robust pipelines for automated data processing and post-processing;

(2) identify and characterize co-binding DNA patterns.

4.1 Paper I. JASPAR 2022: the 9th release of the open-access database of transcription factor binding profiles

Castro-Mondragon JA*, Riudavets-Puig R*, Rauluseviciute I*, Lemma RB, Turchi L, Blanc-Mathieu R, Lucas J, Boddie P, Khan A, Manosalva Pérez N, Fornes O, Leung TY, Aguirre A, Hammal F, Schmelter D, Baranasic D, Ballester B, Sandelin A, Lenhard B, Vandepoele K, Wasserman WW, Parcy F, Mathelier A. *JASPAR 2022: the 9th release of the open-access database of transcription factor binding profiles*. Nucleic Acids Research. 2022 Jan 7;50(D1):D165-73; doi: 10.1093/nar/gkab1113.

Paper I describes the 2022 release of the JASPAR database. JASPAR is a widely used open-access repository of transcription factor binding motifs maintained since 2004. The database is updated regularly with new motifs, features, and tools. One of the biggest strengths of JASPAR is that a team of experts manually curates motifs before they enter the database. We manually curate two types of motifs: (1) in-house *de novo* discovered motifs from genomic data and (2) external motifs from individual papers or databases. We mine genomic data from databases (for example, ReMap [134], and GTRD [135]) or from individual publications. Currently, JASPAR has two motif collections: CORE and UNVALIDATED. The CORE collection contains profiles with orthogonal support.

*These authors should be considered joint first authors.

The UNVALIDATED collection has profiles of the same sound quality but without validation by independent orthogonal experiments.

In 2022, the CORE and UNVALIDATED collections were complemented with 501 and 182 profiles, respectively. The CORE collection increased by 19% compared to the previous release. In addition to expanding the motif collections, we revisited the existing motifs. We revised the UNVALIDATED collection profiles to check for possible new validations that would move the profiles into the CORE collection. In 2022, we upgraded 52 such profiles. We extensively revised the metadata inconsistencies. Metadata fields, such as TF class and family, were updated with the newest available information.

Together with the motif collections, we provide additional features such as motif clustering, familial TF binding motifs, TFBS predictions, and genome browser tracks. We individually clustered CORE collection motifs for each taxon and the CORE and UNVALIDATED collection motifs. Clusters are visualized as radial trees and annotated with TF class information. Furthermore, we used the motif clusters to construct familial binding profiles. A familial binding motif represents the DNA binding preference of closely related TFs. We used the latest versions of the motifs in the CORE collection to scan the most popular genomes to predict TFBSs. Predicted binding sites are available to users along with genome browser tracks. A new addition in 2022 was word clouds, which summarize the information associated with the TFs in related abstracts from the PubMed database of biomedical literature [136]. The JASPAR database has a user-friendly website, but it is also accessible via the RESTful API, R Bioconductor package, and pyJASPAR for advanced users. We provide several computational tools to accompany the data. In this release, we added an enrichment analysis tool that allows users to study the enrichment of the binding of TFs in a set of genomic regions.

My contributions to this paper included experimental and computational TF-DNA interaction data mining from repositories and literature, data processing, manual curation of the motifs and their metadata, and revision of the existing motif metadata. In addition, I worked on the update and development of software components, such as the motif discovery pipeline, to obtain candidate DNA binding motifs. I was involved in the downstream analysis, including computation and revision of matrix clusters. Furthermore, I contributed to updating the JASPAR website content. Finally, my contributions include paper writing and coordination of the international team effort.

4.2 Paper II. JASPAR 2024: 20th anniversary of the open-access database of transcription factor binding profiles

Rauluseviciute I*, Riudavets-Puig R*, Blanc-Mathieu R, Castro-Mondragon JA, Ferenc K, Kumar V, Lemma RB, Lucas J, Chèneby J, Baranasic D, Khan A, Fornes O, Gundersen S, Johansen M, Hovig E, Lenhard B, Sandelin A, Wasserman WW, Parcy F, Mathelier A. *JASPAR 2024: 20th anniversary of the open-access*

4.2. Paper II. JASPAR 2024: 20th anniversary of the open-access database of transcription factor binding profiles

database of transcription factor binding profiles. Nucleic Acids Research. 2024 Jan 5;52(D1):D174-82; doi: 10.1093/nar/gkad1059.

Paper II describes the latest release of the JASPAR database. In 2024, the CORE and UNVALIDATED collections were complemented with 404 and 298 profiles, respectively. We validated and moved 72 profiles to the CORE collection. We expanded the insects and nematodes motif collections by 90% and 140%, respectively. The plant TF classification was extensively revised and is now based on Plant-TFClass [57]. One of the most significant updates in the 2024 release was the application of motif trimming. We trimmed 75% (1,869 out of 2,506) of the JASPAR 2022 collection motifs and all incoming new motifs. We applied an algorithm to remove the low IC positions on the flanks of the motifs to improve the quality of our collections. The trimming procedure will become a standard when preparing the motifs for manual curation in future releases.

For this release, we also provide genomic tracks of predicted TFBSs in 8 organisms, with mouse and human tracks available as native UCSC Genome browser tracks. In addition, we provide updated motif clusters for each taxon, word clouds, and familial TF binding motifs. We added a TFBS extraction tool to JASPAR's featured tools collection for easier extraction of TFBSs in a genomic region of interest.

The JASPAR database update process is complex, and we have refined it into three operations, abbreviated as "3D": **D**iscovery of the motifs, **D**ownstream analysis, and **D**eployment of the website. Each process has its software component as a pipeline. The motif discovery operation finds new motifs in the genomic data and processes them for manual curation. It also processes external motifs found in the literature or databases by visualizing them as sequence logos and comparing them with the current JASPAR motif collections. The downstream analysis prepares manually curated motifs for the website. In addition, the downstream analysis operation performs all additional analyses to produce the data needed for the website. Finally, the deployment workflow collects all produced data and updates the website. All software components received updates to implement new features and increase the reliability and performance of the entire process of JASPAR update.

My contributions to this paper included experimental and computational TF-DNA interaction data mining from repositories and literature, data processing, manual curation of the motifs and their metadata, and revision of the existing motifs and their metadata. I contributed to the software development of motif discovery and downstream analysis pipelines. The motif discovery pipeline was updated to process external motifs along with the discovered ones. I was responsible for motif trimming implementation, application, and comparative analysis of trimmed and untrimmed motifs. In addition, I participated in the curated motif downstream analysis by computing word clouds and familial binding profiles. I contributed to updating the JASPAR website content. Finally, my contributions include paper writing and coordination of the international team effort.

*These authors should be considered joint first authors.

4.3 Paper III. Identification of transcription factor co-binding patterns with non-negative matrix factorization

Rauluseviciute I, Launay T, Barzaghi G, Nikumbh S, Lenhard B, Krebs AR, Castro-Mondragon JA, Mathelier A. *Identification of transcription factor co-binding patterns with non-negative matrix factorization*. bioRxiv. 2023:2023-04.

Paper III describes a non-negative matrix factorization-based method to identify co-binding DNA patterns. The COBIND tool identifies DNA patterns in neighborhoods of the sequences referred to as anchors. We applied COBIND to a collection of transcription factor binding sites to automatically identify potential TF co-binding patterns in flanking regions. Interpreting such predicted DNA patterns can give insights into TF co-binding and the more complex gene regulation mechanism it creates.

In COBIND, NMF is applied to one-hot encoded sequences and pinpoints the enriched DNA patterns at fixed instances. We used 5,699 TFBS datasets from the UniBind database for 401 TFs in seven species [130]. As a proof-of-concept, we present a classical example of POU5F1 co-binding with SOX2 or SOX17. SOX2 and SOX17 TFBS datasets, we were able to recover the DNA pattern that represents the POU5F1 motif with different spacings, corresponding to canonical (SOX2::POU5F1 pair) and compressed (SOX17::POU5F1 pair) motifs. Through motif similarity and protein-protein interaction (PPI) data, we inferred that POU5F1 most likely binds the discovered motif.

We evaluated all discovered co-binding patterns with motif similarity and PPI analysis. 78% (462 out of 591) of the patterns had support in the PPI data. For example, TEAD1 is co-binding with TEAD family members. However, some inferred pairs have not yet been reported in the PPI data or do not have a good motif match. In *A. thaliana* genome, the same co-binding pattern was found for HY5, ABF1, ABF3, ABF4, GBF2, and GBF3 anchors. Motif analysis revealed the similarity with the motif recognized by human NF-Y TFs (NF-YC and NF-YA). PPI data does not currently support these interactions. Another example is a co-binding pattern for CTCF. We found a co-binding pattern with two different spacings in *H. sapiens*, *M. musculus* and *D. rerio* TFBS datasets. The literature reported that another TF does not bind the predicted DNA pattern but CTCF itself. CTCF is a zinc-finger protein, and in a fraction of its binding sites, 9–11 zinc fingers bind to an extended portion of the motif.

We demonstrated that the regions with predicted co-binding patterns are likely functionally relevant as they are more evolutionarily conserved than those with only the anchor TFBSs. Across all predictions, we systematically compared the conservation at the anchor locations and predicted co-binding sites. We observed that 23–83% of the datasets from different species with predicted co-binding patterns showed significantly higher conservation at the co-binding motif locations compared to the alternative locations in regions only with the anchor motifs. For anchor sites, 31–67% of the datasets demonstrated a more conserved anchor site in

4.3. Paper III. Identification of transcription factor co-binding patterns with non-negative matrix factorization

the genomic regions with predicted co-binding patterns. We observed for 23–64% of datasets that the locations of both the anchor and co-binding patterns were more conserved in the co-bound regions. More datasets exhibit increased conservation in the co-bound regions compared to what would be expected by chance.

Furthermore, we investigated regions with and without predicted co-binders by analyzing DHS footprints from matching human cell types. We found that in 85% of co-binding pattern and DHS dataset pairs, DHS footprints were significantly deeper at the locations of the co-binding patterns in regions with predicted co-binding motifs compared to the genomic regions without (expectation by chance was 20%). We observed significantly more profound DHS footprints in anchor locations with predicted co-binding patterns for 78% of the pairs (expectation by chance was 21%). Considering deeper footprints for anchor and co-binding pattern sites together, we observed significance for 73% of the pattern-dataset pairs (zero from random expectation).

Finally, we utilized published single-molecule footprinting (SMF) data to probe the co-occupancy of TFs at the predicted co-binding patterns. SMF data from mouse embryonic stem cells (mESCs) confirmed that the predicted co-binding events associated with a set of TFs are likely occurring on the same DNA molecules. We analyzed predictions for 17 anchor TFs and found co-binding patterns in 1,694 regions. For each of these genomic regions and regions with only anchor motifs, we determined the proportions of molecules in five states: accessible, occupied by nucleosomes, occupied at the anchor motif site, only occupied at the co-binding motif site, or occupied at both sites. Overall, we found that a significantly more significant fraction of molecules was co-occupied at both sites in the regions with COBIND-predicted co-binding motifs compared to the genomic regions only with the anchor motifs. When looking at the individual anchors, significance was observed for 8 out of 17 analyzed anchor TFs.

We introduced the computational framework COBIND (https://bitbucket.org/CBGR/cobind_tool/src/main/) for DNA motif pattern discovery *de novo* in the vicinity of predetermined regions or a set of sequences. During extensive analysis, we produced a collection of predicted co-binding patterns in 7 species, which we provide to the community at https://cbgr.bitbucket.io/COBIND_results_page/. DNA patterns discovered in the fixed locations from known TFBSs reveal the possible cooperative events and can be studied to investigate more complex TF-DNA interactions.

My contributions to this paper included developing the method, performing the analysis, and preparing the COBIND tool and resource with the results. I analyzed and interpreted the results and wrote the paper.

Bibliography

Bibliography

- [1] Dennis W Ross. “The human genome: information content and structure.” In: *Hospital Practice* 34.6 (1999), pp. 49–65.
- [2] Tom Misteli. “The self-organizing genome: principles of genome architecture and function.” In: *Cell* 183.1 (2020), pp. 28–45.
- [3] David Staněk and Archa H Fox. “Nuclear bodies: news insights into structure and function.” In: *Current Opinion in Cell Biology* 46 (2017), pp. 94–101.
- [4] Leslie Pray. “Discovery of DNA structure and function: Watson and Crick.” In: *Nature Education* 1.1 (2008), p. 100.
- [5] Ralf Dahm. “Discovering DNA: Friedrich Miescher and the early years of nucleic acid research.” In: *Human genetics* 122 (2008), pp. 565–581.
- [6] James D Watson and Francis HC Crick. “Molecular structure of nucleic acids: a structure for deoxyribose nucleic acid.” In: *Nature* 171.4356 (1953), pp. 737–738.
- [7] Andrew Travers and Georgi Muskhelishvili. “DNA structure and function.” In: *The FEBS Journal* 282.12 (2015), pp. 2279–2295.
- [8] Erwin Chargaff et al. “The composition of the desoxyribonucleic acid of salmon sperm.” In: *Journal of Biological Chemistry* 192.1 (1951), pp. 223–230.
- [9] Carl O. Pabo and Robert T. Sauer. “Protein-DNA recognition.” In: *Annual Review of Biochemistry* 53.1 (1984). PMID: 6236744, pp. 293–321.
- [10] Jennifer M Luppino et al. “Cohesin promotes stochastic domain intermingling to ensure proper regulation of boundary-proximal genes.” In: *Nature genetics* 52.8 (2020), pp. 840–848.
- [11] Olivia Morrison and Jitendra Thakur. “Molecular complexes at euchromatin, heterochromatin and centromeric chromatin.” In: *International Journal of Molecular Sciences* 22.13 (2021), p. 6922.
- [12] Yuntao S Mao, Bin Zhang, and David L Spector. “Biogenesis and function of nuclear bodies.” In: *Trends in Genetics* 27.8 (2011), pp. 295–306.
- [13] Leonardo Marino-Ramirez et al. “Histone structure and nucleosome stability.” In: *Expert review of proteomics* 2.5 (2005), pp. 719–729.
- [14] Paul B Talbert and Steven Henikoff. “Histone variants at a glance.” In: *Journal of Cell Science* 134.6 (2021), jcs244749.

Bibliography

- [15] Magdalena A Karpinska and Aukje Marieke Oudelaar. “The role of loop extrusion in enhancer-mediated gene activation.” In: *Current Opinion in Genetics & Development* 79 (2023), p. 102022.
- [16] Paul G Higgs. “RNA secondary structure: physical and computational aspects.” In: *Quarterly reviews of biophysics* 33.3 (2000), pp. 199–253.
- [17] Ann-Bin Shyu, Miles F Wilkinson, and Ambro van Hoof. “Messenger RNA regulation: to translate or to degrade.” In: *The EMBO Journal* 27.3 (2008), pp. 471–481.
- [18] Medha Raina and Michael Ibba. “tRNAs as regulators of biological processes.” In: *Frontiers in genetics* 5 (2014), p. 171.
- [19] Melissa J Moore. “From birth to death: the complex lives of eukaryotic mRNAs.” In: *Science* 309.5740 (2005), pp. 1514–1518.
- [20] Yeon Lee and Donald C Rio. “Mechanisms and regulation of alternative pre-mRNA splicing.” In: *Annual review of biochemistry* 84 (2015), pp. 291–323.
- [21] Nuno André Faustino and Thomas A Cooper. “Pre-mRNA splicing and human disease.” In: *Genes & development* 17.4 (2003), pp. 419–437.
- [22] Lin Xiao et al. “Disorders and roles of tsRNA, snoRNA, snRNA and piRNA in cancer.” In: *Journal of Medical Genetics* 59.7 (2022), pp. 623–631.
- [23] Minna U Kaikkonen, Michael TY Lam, and Christopher K Glass. “Non-coding RNAs as regulators of gene expression and epigenetics.” In: *Cardiovascular research* 90.3 (2011), pp. 430–440.
- [24] DB Nikolov and SK Burley. “RNA polymerase II transcription initiation: a structural view.” In: *Proceedings of the National Academy of Sciences* 94.1 (1997), pp. 15–22.
- [25] Xin Liu, David A Bushnell, and Roger D Kornberg. “RNA polymerase II transcription: structure and mechanism.” In: *Biochimica et Biophysica Acta (BBA)-Gene Regulatory Mechanisms* 1829.1 (2013), pp. 2–8.
- [26] Wyeth W Wasserman and Albin Sandelin. “Applied bioinformatics for the identification of regulatory elements.” In: *Nature Reviews Genetics* 5.4 (2004), pp. 276–287.
- [27] Vanja Haberle and Boris Lenhard. “Promoter architectures and developmental gene regulation.” In: *Seminars in cell & developmental biology*. Vol. 57. Elsevier. 2016, pp. 11–23.
- [28] James T Kadonaga. “Perspectives on the RNA polymerase II core promoter.” In: *Wiley Interdisciplinary Reviews: Developmental Biology* 1.1 (2012), pp. 40–51.
- [29] Liliana Villao-Uzho et al. “Plant promoters: their identification, characterization, and role in gene regulation.” In: *Genes* 14.6 (2023), p. 1226.

- [30] Serge Saxonov, Paul Berg, and Douglas L Brutlag. “A genome-wide analysis of CpG dinucleotides in the human genome distinguishes two distinct classes of promoters.” In: *Proceedings of the National Academy of Sciences* 103.5 (2006), pp. 1412–1417.
- [31] Marta Kulis and Manel Esteller. “DNA methylation and cancer.” In: *Advances in genetics* 70 (2010), pp. 27–56.
- [32] Aimée M Deaton and Adrian Bird. “CpG islands and the regulation of transcription.” In: *Genes & development* 25.10 (2011), pp. 1010–1022.
- [33] Edwin Smith and Ali Shilatifard. “Enhancer biology and enhanceropathies.” In: *Nature structural & molecular biology* 21.3 (2014), pp. 210–219.
- [34] Len A Pennacchio et al. “Enhancers: five essential questions.” In: *Nature Reviews Genetics* 14.4 (2013), pp. 288–295.
- [35] Edgar Serfling, Maria Jasin, and Walter Schaffner. “Enhancers and eukaryotic gene transcription.” In: *Trends in Genetics* 1 (1985), pp. 224–230.
- [36] Andrew Field and Karen Adelman. “Evaluating enhancer function and transcription.” In: *Annual review of biochemistry* 89 (2020), pp. 213–234.
- [37] François Spitz and Eileen EM Furlong. “Transcription factors: from enhancer binding to developmental control.” In: *Nature reviews genetics* 13.9 (2012), pp. 613–626.
- [38] Manqing Wu and Jun Shen. “From super-enhancer non-coding RNA to immune checkpoint: frameworks to functions.” In: *Frontiers in oncology* 9 (2019), p. 1307.
- [39] Nathaniel D Tippens, Anniina Vihervaara, and John T Lis. “Enhancer transcription: what, where, when, and why?” In: *Genes & development* 32.1 (2018), pp. 1–3.
- [40] Iain F Davidson and Jan-Michael Peters. “Genome folding through loop extrusion by SMC complexes.” In: *Nature reviews Molecular cell biology* 22.7 (2021), pp. 445–464.
- [41] Orsolya Symmons et al. “Functional and topological characteristics of mammalian regulatory domains.” In: *Genome research* 24.3 (2014), pp. 390–400.
- [42] Baoxu Pang and Michael P Snyder. “Systematic identification of silencers in human cells.” In: *Nature genetics* 52.3 (2020), pp. 254–263.
- [43] Baoxu Pang et al. “Identification of non-coding silencer elements and their regulation of gene expression.” In: *Nature Reviews Molecular Cell Biology* 24.6 (2023), pp. 383–395.
- [44] David B Beck et al. “PR-Set7 and H4K20me1: at the crossroads of genome integrity, cell cycle, chromosome condensation, and transcription.” In: *Genes & development* 26.4 (2012), pp. 325–337.

Bibliography

- [45] Thanasis Mitsis et al. “Transcription factors and evolution: an integral part of gene expression.” In: *World Academy of Sciences Journal* 2.1 (2020), pp. 3–8.
- [46] Paula Weidemüller et al. “Transcription factors: Bridge between cell signaling and gene regulation.” In: *Proteomics* 21.23-24 (2021), p. 2000034.
- [47] Samuel A Lambert et al. “The human transcription factors.” In: *Cell* 172.4 (2018), pp. 650–665.
- [48] Edgar Wingender, Torsten Schoeps, and Jürgen Dönitz. “TFClass: an expandable hierarchical classification of human transcription factors.” In: *Nucleic acids research* 41.D1 (2013), pp. D165–D170.
- [49] Ling Zhu and Enamul Huq. “Mapping functional domains of transcription factors.” In: *Plant Transcription Factors: Methods and Protocols* (2011), pp. 167–184.
- [50] S.D. Minchin and S.J.W. Busby. “Transcription Factors.” In: (2013). Ed. by Stanley Maloy and Kelly Hughes, pp. 93–96.
- [51] Edgar Wingender. “Criteria for an updated classification of human transcription factor DNA-binding domains.” In: *Journal of Bioinformatics and Computational Biology* 11.01 (2013), p. 1340007.
- [52] Dhanusha Yesudhas et al. “Proteins recognizing DNA: Structural uniqueness and versatility of DNA-binding domains in stem cell transcription factors.” In: *Genes* 8.8 (2017), p. 192.
- [53] Helen M Berman et al. “The protein data bank.” In: *Nucleic acids research* 28.1 (2000), pp. 235–242.
- [54] Juan M Vaquerizas et al. “A census of human transcription factors: function, expression and evolution.” In: *Nature Reviews Genetics* 10.4 (2009), pp. 252–263.
- [55] Edgar Wingender et al. “TFClass: expanding the classification of human transcription factors to their mammalian orthologs.” In: *Nucleic acids research* 46.D1 (2018), pp. D343–D347.
- [56] Edgar Wingender et al. “TFClass: a classification of human transcription factors and their rodent orthologs.” In: *Nucleic acids research* 43.D1 (2015), pp. D97–D102.
- [57] Romain Blanc-Mathieu et al. “Plant-TFClass: a structural classification for plant transcription factors.” In: *Trends in Plant Science* 29.1 (2024), pp. 40–51.
- [58] Rezvan Ehsani, Shahram Bahrami, and Finn Drabløs. “Feature-based classification of human transcription factors into hypothetical sub-classes related to regulatory function.” In: *BMC bioinformatics* 17 (2016), pp. 1–16.
- [59] Amandine Barral and Kenneth S Zaret. “Pioneer factors: roles and their regulation in development.” In: *Trends in Genetics* (2023).

- [60] Hannah K Long, Sara L Prescott, and Joanna Wysocka. “Ever-changing landscapes: transcriptional enhancers in development and evolution.” In: *Cell* 167.5 (2016), pp. 1170–1187.
- [61] Richard I Sherwood et al. “Discovery of directional and nondirectional pioneer transcription factors by modeling DNase profile magnitude and shape.” In: *Nature biotechnology* 32.2 (2014), pp. 171–178.
- [62] Kenneth S Zaret and Jason S Carroll. “Pioneer transcription factors: establishing competence for gene expression.” In: *Genes & development* 25.21 (2011), pp. 2227–2241.
- [63] Lisa Ann Cirillo et al. “Opening of compacted chromatin by early developmental transcription factors HNF3 (FoxA) and GATA-4.” In: *Molecular cell* 9.2 (2002), pp. 279–289.
- [64] Xinyang Yu and Michael J Buck. “Pioneer factors and their in vitro identification methods.” In: *Molecular Genetics and Genomics* 295 (2020), pp. 825–835.
- [65] Valentina Boeva. “Analysis of genomic sequence motifs for deciphering transcription factor binding and transcriptional regulation in eukaryotic cells.” In: *Frontiers in genetics* 7 (2016), p. 24.
- [66] Roza Berhanu Lemma et al. “Pioneer transcription factors are associated with the modulation of DNA methylation patterns across cancers.” In: *Epigenetics & chromatin* 15.1 (2022), p. 13.
- [67] Giovanna Ambrosini et al. “Insights gained from a comprehensive all-against-all transcription factor binding motif benchmarking study.” In: *Genome biology* 21 (2020), pp. 1–18.
- [68] Hugo G Schmidt et al. “An integrated model of transcription factor diffusion shows the importance of intersegmental transfer and quaternary protein structure for target site finding.” In: *PLOS one* 9.10 (2014), e108575.
- [69] Sachi Inukai, Kian Hong Kock, and Martha L Bulyk. “Transcription factor–DNA binding: beyond binding site motifs.” In: *Current opinion in genetics & development* 43 (2017), pp. 110–119.
- [70] Raluca Gordân, Alexander J Hartemink, and Martha L Bulyk. “Distinguishing direct versus indirect transcription factor–DNA interactions.” In: *Genome research* 19.11 (2009), pp. 2090–2100.
- [71] Žiga Avsec et al. “Base-resolution models of transcription-factor binding reveal soft motif syntax.” In: *Nature genetics* 53.3 (2021), pp. 354–366.
- [72] Ilias Georgakopoulos-Soares et al. “Transcription factor binding site orientation and order are major drivers of gene regulatory activity.” In: *Nature communications* 14.1 (2023), p. 2333.
- [73] Mengchi Wang et al. “Identification of DNA motifs that regulate DNA methylation.” In: *Nucleic Acids Research* 47.13 (2019), pp. 6753–6768.
- [74] Dirk Schübeler. “Function and information content of DNA methylation.” In: *Nature* 517.7534 (2015), pp. 321–326.

Bibliography

- [75] Madeleine P Ball et al. “Targeted and genome-scale strategies reveal gene-body methylation signatures in human cells.” In: *Nature biotechnology* 27.4 (2009), pp. 361–368.
- [76] Yimeng Yin et al. “Impact of cytosine methylation on DNA binding specificities of human transcription factors.” In: *Science* 356.6337 (2017), eaaaj2239.
- [77] Jan Grau, Florian Schmidt, and Marcel H Schulz. “Widespread effects of DNA methylation and intra-motif dependencies revealed by novel transcription factor binding models.” In: *Nucleic Acids Research* 51.18 (2023), e95–e95.
- [78] Coby Viner et al. “Modeling methyl-sensitive transcription factor motifs with an expanded epigenetic alphabet.” In: *Genome biology* 25.1 (2024), p. 11.
- [79] Marcel Geertz and Sebastian J Maerkl. “Experimental strategies for studying transcription factor–DNA binding specificities.” In: *Briefings in functional genomics* 9.5-6 (2010), pp. 362–373.
- [80] Kenneth W Kinzler and Bert Vogelstein. “The GLI gene encodes a nuclear protein which binds specific sequences in the human genome.” In: *Molecular and cellular biology* 10.2 (1990), pp. 634–642.
- [81] Arttu Jolma et al. “Multiplexed massively parallel SELEX for characterization of human transcription factor binding specificities.” In: *Genome research* 20.6 (2010), pp. 861–873.
- [82] Arttu Jolma et al. “DNA-dependent formation of transcription factor pairs alters their binding specificity.” In: *Nature* 527.7578 (2015), pp. 384–388.
- [83] Michael F Berger and Martha L Bulyk. “Protein binding microarrays (PBMs) for rapid, high-throughput characterization of the sequence specificities of DNA binding proteins.” In: *Gene Mapping, Discovery, and Expression: Methods and Protocols* (2006), pp. 245–260.
- [84] Ronan C O’Malley et al. “Cistrome and epicistrome features shape the regulatory DNA landscape.” In: *Cell* 165.5 (2016), pp. 1280–1292.
- [85] Anna Bartlett et al. “Mapping genome-wide transcription-factor binding sites using DAP-seq.” In: *Nature protocols* 12.8 (2017), pp. 1659–1672.
- [86] David S Johnson et al. “Genome-wide mapping of in vivo protein-DNA interactions.” In: *Science* 316.5830 (2007), pp. 1497–1502.
- [87] Peter J Park. “ChIP-seq: advantages and challenges of a maturing technology.” In: *Nature reviews genetics* 10.10 (2009), pp. 669–680.
- [88] Ho Sung Rhee and B Franklin Pugh. “Comprehensive genome-wide protein-DNA interactions detected at single-nucleotide resolution.” In: *Cell* 147.6 (2011), pp. 1408–1419.
- [89] Matthew J Rossi, William KM Lai, and B Franklin Pugh. “Simplified ChIP-exo assays.” In: *Nature communications* 9.1 (2018), p. 2842.

- [90] Rene Welch et al. “Data exploration, quality control and statistical analysis of ChIP-exo/nexus experiments.” In: *Nucleic Acids Research* 45.15 (2017), e145–e145.
- [91] Ssu-Yu Yeh and Ho Sung Rhee. “The ChIP-Exo Method to Identify Genomic Locations of DNA-Binding Proteins at Near Single Base-Pair Resolution.” In: *DNA-Protein Interactions: Methods and Protocols*. Springer, 2022, pp. 33–48.
- [92] Hatice S Kaya-Okur et al. “CUT&Tag for efficient epigenomic profiling of small samples and single cells.” In: *Nature communications* 10.1 (2019), p. 1930.
- [93] Hatice S Kaya-Okur et al. “Efficient low-cost chromatin profiling with CUT&Tag.” In: *Nature protocols* 15.10 (2020), pp. 3264–3283.
- [94] Peter J Skene and Steven Henikoff. “An efficient targeted nuclease strategy for high-resolution mapping of DNA binding sites.” In: *elife* 6 (2017), e21856.
- [95] Lingyun Song and Gregory E Crawford. “DNase-seq: a high-resolution technique for mapping active gene regulatory elements across the genome from mammalian cells.” In: *Cold Spring Harbor Protocols* 2010.2 (2010), pdb-prot5384.
- [96] Robert E Thurman et al. “The accessible chromatin landscape of the human genome.” In: *Nature* 489.7414 (2012), pp. 75–82.
- [97] Feng Yan et al. “From reads to insight: a hitchhiker’s guide to ATAC-seq data analysis.” In: *Genome biology* 21 (2020), pp. 1–16.
- [98] Can Sönmezer et al. “Molecular co-occupancy identifies transcription factor binding cooperativity in vivo.” In: *Molecular cell* 81.2 (2021), pp. 255–267.
- [99] Rozemarijn WD Kleinendorst et al. “Genome-wide quantification of transcription factor binding at single-DNA-molecule resolution using methyl-transferase footprinting.” In: *Nature protocols* 16.12 (2021), pp. 5673–5706.
- [100] Ieva Rauluseviciute et al. “Identification of transcription factor co-binding patterns with non-negative matrix factorization.” In: *bioRxiv* (2023), pp. 2023–04.
- [101] Matthew T Weirauch et al. “Evaluation of methods for modeling transcription factor sequence specificity.” In: *Nature biotechnology* 31.2 (2013), pp. 126–134.
- [102] Gary D Stormo. “Modeling the specificity of protein-DNA interactions.” In: *Quantitative biology* 1.2 (2013), pp. 115–130.
- [103] Giovanna Ambrosini, Romain Groux, and Philipp Bucher. “PWMScan: a fast tool for scanning entire genomes with a position-specific weight matrix.” In: *Bioinformatics* 34.14 (2018), pp. 2483–2484.
- [104] Anthony Mathelier and Wyeth W Wasserman. “The next generation of transcription factor binding site prediction.” In: *PLoS computational biology* 9.9 (2013), e1003214.

Bibliography

- [105] Yue Zhao et al. “Improved models for transcription factor binding site identification using nonindependent interactions.” In: *Genetics* 191.3 (2012), pp. 781–790.
- [106] Anthony Mathelier et al. “DNA shape features improve transcription factor binding site predictions in vivo.” In: *Cell systems* 3.3 (2016), pp. 278–286.
- [107] Tianyin Zhou et al. “DNASHape: a method for the high-throughput prediction of DNA structural features on a genomic scale.” In: *Nucleic acids research* 41.W1 (2013), W56–W62.
- [108] Zixuan Wang et al. “Towards a better understanding of TF-DNA binding prediction from genomic features.” In: *Computers in Biology and Medicine* 149 (2022), p. 105993.
- [109] Avanti Shrikumar et al. “Technical note on transcription factor motif discovery from importance scores (TF-MoDISco) version 0.5. 6.5.” In: *arXiv preprint arXiv:1811.00416* (2018).
- [110] Timothy L Bailey, Charles Elkan, et al. “Fitting a mixture model by expectation maximization to discover motifs in bipolymers.” In: (1994).
- [111] Timothy L Bailey. “STREME: accurate and versatile sequence motif discovery.” In: *Bioinformatics* 37.18 (2021), pp. 2834–2840.
- [112] Morgane Thomas-Chollier et al. “RSAT 2011: regulatory sequence analysis tools.” In: *Nucleic acids research* 39.suppl_2 (2011), W86–W91.
- [113] Morgane Thomas-Chollier et al. “RSAT peak-motifs: motif analysis in full-size ChIP-seq datasets.” In: *Nucleic acids research* 40.4 (2012), e31–e31.
- [114] Walter Santana-Garcia et al. “RSAT 2022: regulatory sequence analysis tools.” In: *Nucleic Acids Research* 50.W1 (2022), W670–W676.
- [115] Jacques van Helden, Bruno André, and Julio Collado-Vides. “Extracting regulatory sites from the upstream region of yeast genes by computational analysis of oligonucleotide frequencies.” In: *Journal of molecular biology* 281.5 (1998), pp. 827–842.
- [116] Jacques van Helden, Marcel li del Olmo, and Jose E Perez-Ortin. “Statistical analysis of yeast genomic downstream sequences reveals putative polyadenylation signals.” In: *Nucleic acids research* 28.4 (2000), pp. 1000–1010.
- [117] Jacques van Helden, Alma F Rios, and Julio Collado-Vides. “Discovering regulatory elements in non-coding sequences by analysis of spaced dyads.” In: *Nucleic acids research* 28.8 (2000), pp. 1808–1818.
- [118] Sven Heinz et al. “Simple combinations of lineage-determining transcription factors prime cis-regulatory elements required for macrophage and B cell identities.” In: *Molecular cell* 38.4 (2010), pp. 576–589.
- [119] Genevieve L Stein-O’Brien et al. “Enter the matrix: factorization uncovers knowledge from omics.” In: *Trends in Genetics* 34.10 (2018), pp. 790–805.

- [120] Sarvesh Nikumbh and Boris Lenhard. “Identifying promoter sequence architectures via a chunking-based algorithm using non-negative matrix factorisation.” In: *PLOS Computational Biology* 19.11 (2023), e1011491.
- [121] Jean-Philippe Brunet et al. “Metagenes and molecular pattern discovery using matrix factorization.” In: *Proceedings of the national academy of sciences* 101.12 (2004), pp. 4164–4169.
- [122] Ieva Rauluseviciute et al. “JASPAR 2024: 20th anniversary of the open-access database of transcription factor binding profiles.” In: *Nucleic Acids Research* 52.D1 (2024), pp. D174–D182.
- [123] Volker Matys et al. “TRANSFAC® and its module TRANSCompel®: transcriptional gene regulation in eukaryotes.” In: *Nucleic acids research* 34.suppl_1 (2006), pp. D108–D110.
- [124] Edgar Wingender. “The TRANSFAC project as an example of framework technology that supports the analysis of genomic regulation.” In: *Briefings in bioinformatics* 9.4 (2008), pp. 326–332.
- [125] Ilya E Vorontsov et al. “HOCOMOCO in 2024: a rebuild of the curated collection of binding models for human and mouse transcription factors.” In: *Nucleic Acids Research* 52.D1 (2024), pp. D154–D163.
- [126] Matthew T Weirauch et al. “Determination and inference of eukaryotic transcription factor sequence specificity.” In: *Cell* 158.6 (2014), pp. 1431–1443.
- [127] Matthew Dyer et al. “MethMotif. Org 2024: a database integrating context-specific transcription factor-binding motifs with DNA methylation patterns.” In: *Nucleic Acids Research* 52.D1 (2024), pp. D222–D228.
- [128] Charles E Grant, Timothy L Bailey, and William Stafford Noble. “FIMO: scanning for occurrences of a given motif.” In: *Bioinformatics* 27.7 (2011), pp. 1017–1018.
- [129] Terrence S Furey. “ChIP-seq and beyond: new and improved methodologies to detect and characterize protein–DNA interactions.” In: *Nature Reviews Genetics* 13.12 (2012), pp. 840–852.
- [130] Rafael Riudavets Puig et al. “UniBind: maps of high-confidence direct TF–DNA interactions across nine species.” In: *BMC genomics* 22.1 (2021), pp. 1–17.
- [131] Marius Gheorghe et al. “A map of direct TF–DNA interactions in the human genome.” In: *Nucleic acids research* 47.4 (2019), e21–e21.
- [132] Shobhit Gupta et al. “Quantifying similarity between motifs.” In: *Genome biology* 8 (2007), pp. 1–9.
- [133] Tom Whittington et al. “Inferring transcription factor complexes from ChIP-seq data.” In: *Nucleic acids research* 39.15 (2011), e98–e98.

Bibliography

- [134] Fayrouz Hammal et al. “ReMap 2022: a database of Human, Mouse, Drosophila and Arabidopsis regulatory regions from an integrative analysis of DNA-binding sequencing experiments.” In: *Nucleic acids research* 50.D1 (2022), pp. D316–D325.
- [135] Semyon Kolmykov et al. “GTRD: an integrated view of transcription regulation.” In: *Nucleic acids research* 49.D1 (2021), pp. D104–D111.
- [136] Kathi Canese and Sarah Weis. “PubMed: the bibliographic database.” In: *The NCBI handbook* 2.1 (2013).
- [137] Matthew Slattery et al. “Absence of a simple code: how transcription factors read the genome.” In: *Trends in biochemical sciences* 39.9 (2014), pp. 381–399.
- [138] Camille Berthelot et al. “Complexity and conservation of regulatory landscapes underlie evolutionary resilience of mammalian gene expression.” In: *Nature ecology & evolution* 2.1 (2018), pp. 152–163.
- [139] Klara Stefflova et al. “Cooperativity and rapid evolution of cobound transcription factors in closely related mammals.” In: *Cell* 154.3 (2013), pp. 530–540.
- [140] Mo Li and Juan Carlos Izpisua Belmonte. “Deconstructing the pluripotency gene regulatory network.” In: *Nature cell biology* 20.4 (2018), pp. 382–392.
- [141] Irene Aksoy et al. “Oct4 switches partnering from Sox2 to Sox17 to reinterpret the enhancer code and specify endoderm.” In: *The EMBO journal* 32.7 (2013), pp. 938–953.
- [142] Victor Levitsky et al. “A single ChIP-seq dataset is sufficient for comprehensive analysis of motifs co-occurrence with MCOT package.” In: *Nucleic acids research* 47.21 (2019), e139–e139.
- [143] Amber W Wang et al. “The dynamic chromatin architecture of the regenerating liver.” In: *Cellular and molecular gastroenterology and hepatology* 9.1 (2020), pp. 121–143.
- [144] Alan P Boyle et al. “High-resolution genome-wide in vivo footprinting of diverse transcription factors in human cells.” In: *Genome research* 21.3 (2011), pp. 456–464.
- [145] Widia Soochit et al. “CTCF chromatin residence time controls three-dimensional genome organization, gene expression and DNA methylation in pluripotent cells.” In: *Nature cell biology* 23.8 (2021), pp. 881–893.
- [146] Quy Xiao Xuan Lin et al. “TFregulomeR reveals transcription factors’ context-specific features and functions.” In: *Nucleic acids research* 48.2 (2020), e10–e10.
- [147] Timothy L Bailey and Philip Machanick. “Inferring direct DNA binding from ChIP-seq.” In: *Nucleic acids research* 40.17 (2012), e128–e128.
- [148] Mark D Wilkinson et al. “The FAIR Guiding Principles for scientific data management and stewardship.” In: *Scientific data* 3.1 (2016), pp. 1–9.

- [149] Leho Tedersoo et al. “Data sharing practices and data availability upon request differ across scientific disciplines.” In: *Scientific data* 8.1 (2021), p. 192.
- [150] Dan L Longo and Jeffrey M Drazen. *Data sharing*. 2016.
- [151] US DOE Joint Genome Institute: Hawkins Trevor 4 Branscomb Elbert 4 Predki Paul 4 Richardson Paul 4 Wenning Sarah 4 Slezak Tom 4 Doggett Norman 4 Cheng Jan-Fang 4 Olsen Anne 4 Lucas Susan 4 Elkin Christopher 4 Uberbacher Edward 4 Frazier Marvin 4 et al. “Initial sequencing and analysis of the human genome.” In: *nature* 409.6822 (2001), pp. 860–921.
- [152] ENCODE Project Consortium et al. “An integrated encyclopedia of DNA elements in the human genome.” In: *Nature* 489.7414 (2012), p. 57.
- [153] European Organization For Nuclear Research and OpenAIRE. *Zenodo*. en. 2013. DOI: [10.25495/7G XK-RD71](https://doi.org/10.25495/7G XK-RD71). URL: <https://www.zenodo.org/>.
- [154] Jatinder Singh. “FigShare.” In: *Journal of Pharmacology and Pharmacotherapeutics* 2.2 (2011), pp. 138–138.
- [155] Jeff Vierstra et al. “Global reference mapping of human transcription factor footprints.” In: *Nature* 583.7818 (2020), pp. 729–736.
- [156] Marion Gremse et al. “The BRENDA Tissue Ontology (BTO): the first all-integrating ontology of all organisms for enzyme sources.” In: *Nucleic acids research* 39.suppl_1 (2010), pp. D507–D513.
- [157] Robert C Gentleman et al. “Bioconductor: open software development for computational biology and bioinformatics.” In: *Genome biology* 5 (2004), pp. 1–16.
- [158] R Core Team. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. Vienna, Austria, 2023. URL: <https://www.R-project.org/>.
- [159] Guido Van Rossum and Fred L. Drake. *Python 3 Reference Manual*. Scotts Valley, CA: CreateSpace, 2009. ISBN: 1441412697.
- [160] Paolo Di Tommaso et al. “Nextflow enables reproducible computational workflows.” In: *Nature biotechnology* 35.4 (2017), pp. 316–319.
- [161] Felix Mölder et al. “Sustainable data analysis with Snakemake.” In: *F1000Research* 10 (2021).
- [162] *Anaconda Software Distribution*. Version Vers. 2-2.4.0. 2020. URL: <https://docs.anaconda.com/>.
- [163] Gregory M Kurtzer, Vanessa Sochat, and Michael W Bauer. “Singularity: Scientific containers for mobility of compute.” In: *PloS one* 12.5 (2017), e0177459.
- [164] Dirk Merkel. “Docker: lightweight linux containers for consistent development and deployment.” In: *Linux journal* 2014.239 (2014), p. 2.
- [165] Monya Baker. “1,500 scientists lift the lid on reproducibility.” In: *Nature* 533.7604 (2016).

Bibliography